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Herbivorous insects independently evolved salivary effectors to regulate plant immunity by destabilizing the malectin-LRR RLP NtRLP4

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eLife Assessment

This study provides an **important** contribution by showing that whiteflies and planthoppers use salivary effectors to suppress plant immunity through the receptor-like protein RLP4, suggesting convergent evolution in these insect lineages. The topic is of clear interest for understanding plant-insect interactions and offers ideas that could stimulate further research in the field. The authors provide **convincing** evidence for the functional roles of the salivary effectors.

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Abstract

Plants utilize on receptor-like proteins and receptor-like kinases (RLPs/RLKs) to perceive and respond to a wide variety of invading pathogens and insect herbivores. While the strategies employed by microbial pathogens to suppress plant immunity have been well characterized, it remains unclear how herbivorous insects counteract receptor-mediated defenses. Here, we show that salivary effectors evolve independently in whiteflies and planthoppers to dampen RLP4-mediated plant immunity. RLP4, as a leucine-rich repeat RLP (LRR-RLP), confers plant resistance against herbivorous insects by forming the RLP4/SOBIR1 complexes. In the whitefly *Bemisia tabaci*, BtRDP, the Aleyrodidae-specific salivary sheath protein, interacts with RLP4 from multiple plant species and promotes its ubiquitin-dependent degradation. Overexpression of NtRLP4 in transgenic plants exerts a detrimental effect on *B. tabaci* by exploiting the crosstalk between the salicylic acid (SA) and jasmonic acid (JA) pathways. Conversely, overexpression of BtRDP or silencing of NtRLP4 effectively alleviates such negative effects. In planthopper *Nilaparvata lugens*, the Delphacidae-restricted salivary protein NISP104 also targets and promotes the degradation of OsRLP4 from rice plants. These findings reveal convergent evolution of salivary proteins in insects, and underscore the complex interactions between plants and herbivorous insects.

Introduction

Plants have evolved sophisticated defense systems to withstand continuous threats from microbial pathogens and herbivorous insects. These defenses function at multiple levels, encompassing physical barriers such as the cuticle and cell wall, chemical defenses that include toxic secondary metabolites and anti-nutritive compounds, and indirect defenses involving the emission of herbivore-induced volatiles to attract natural enemies of the herbivores [1, 2]. Beyond these general strategies, plants also rely on the highly specialized molecular immune responses to detect

and rapidly respond to invaders. The activation of innate immunity by membrane-localized receptors is a highly conserved mechanism among eukaryotes [3]. In sessile plants, the first line of defense against potential pathogenic microbes or insects relies on cell surface pattern-recognition receptors (PRRs) [4]. The plant genome can encode hundreds of PRRs, including receptor-like kinases (RLKs) and receptor-like proteins (RLPs) [4]. RLKs are consisted of different types of N-terminal ectodomains, a single transmembrane domain, and a cytoplasmic protein kinase domain, while RLPs possess a similar structure but lack the kinase domain [5, 6]. As RLPs lack the intracellular signaling domains, they are anticipated to associate with adaptor kinases to form the bimolecular receptor kinases. For example, suppressor of BAK1-interacting receptor-like kinase 1 (SOBIR1) is reported to act as a common adaptor for most, if not all, of the leucine-rich repeat RLP (LRR-RLP) [6]. Despite the large number of PRRs encoded by plant genomes, only a few of them have been well characterized for their functions in recognizing microbial-associated molecular patterns (MAMPs), damage-associated molecular patterns (DAMPs), and herbivore-associated molecular patterns (HAMPs) [5, 7]. For instance, FLS2 is known to recognize bacterial flagellin [8], EFR can recognize the bacterial elongation factor Tu [9], and EIX2 is responsible for the recognition of the fungal ethylene-inducing xylanase [10].

Piercing-sucking insects, such as whiteflies, aphids, and planthoppers, can damage plants by feeding or transmitting viruses. They inject a mixture of saliva into plant tissues, which enhances their feeding efficiency by binding to leaked calcium, degrading extracellular DNA, and regulating plant hormonal signals [11–14]. In response, plants have evolved the ability to detect herbivores by sensing HAMPs present in insect saliva [7, 15]. This triggers pattern-triggered immunity (PTI), including mitogen-activated protein kinase (MAPK) cascades, reactive oxygen species (ROS), and hormone signaling [16–18]. A few salivary proteins have been reported to activate plant PTI, but the specific PRRs that recognize these proteins and trigger plant immunity remain largely unexplored [19–21]. One exception is the inceptin receptor (INR), which is a legume-specific LRR-RLP that can recognize the Vu-In in insect oral secretions [7]. Moreover, plant PRRs have been extensively reported to be targeted by pathogen effectors [22–24]. However, it remains largely unknown how insects cope with plant PRRs.

The whitefly *Bemisia tabaci* (Hemiptera: Aleyrodidae) and brown planthopper *Nilaparvata lugens* (Hemiptera: Delphacidae) are notorious pests globally. During their feeding process, both insects secrete abundant species-specific salivary proteins into host plants [25, 26]. It is reported that plants initiate defense responses upon the recognition of insect feeding [19, 27]. However, the specific PRRs responsible for this process are still unknown. Both insects can successfully ingest phloem saps with limited plant defenses. At present, it remains unknown how insects attenuate plant defenses. Given that numerous defense mechanisms are conserved across plants [3, 28], it is interesting to investigate whether insect evolves a similar strategy to aid feeding during convergent evolution. In this study, the functions of RLPs in the plant immunity responsive to piercing-sucking insects were investigated, and *B. tabaci* and *N. lugens* were demonstrated to independently evolve salivary proteins to regulate the post-translational modification of RLPs.

Results

BtRDP is most abundantly expressed in salivary glands and secreted into host plants

Based on the transcriptomic analysis, BtRDP (*B. tabaci* RLP-degrading protein; GenBank accession: MN738093) was identified as the most abundantly expressed gene in salivary glands [26]. It was a secretory protein featuring an N-terminal signal peptide at the N-terminal, and 6 unique peptides from BtRDP were detected in the *B. tabaci* watery saliva (Fig. S1). As there was no gene homolog identified in the NCBI nr database, BtRDP was subjected to BLAST analysis against the transcriptomic or genomic data of 30 insect species. RDP was distributed in all the analyzed Aleyrodidae species, while no RDP homolog was detected in other species (Table S1). Phylogenetic analysis indicated that RDPs from five *B. tabaci* cryptic species were clustered in the same clad (Fig. S2).

A group of *B. tabaci* were allowed to feed on *Nicotiana tabacum* plants for 24 h, and the presence of BtRDP in the infested and non-infested samples were assayed. One band between 26–33 kDa was detected in the *N. tabacum* plants infested by *B. tabaci*, but not in the non-infested plants (Fig. 1a). This band matched the molecular weight of BtRDP identified in the salivary glands, suggesting BtRDP as a salivary protein. According to Western-blotting, qPCR and transcriptomic analysis, BtRDP was nearly exclusively expressed in the salivary glands (Fig. 1b, c; Fig. S3a; Fig. S4). In addition, BtRDP was highly expressed in the adult stages, but lowly expressed in the egg, nymph, and pseudopupa stages (Fig. 1d; Fig. S3b). According to immunohistochemical (IHC) staining, BtRDP was specifically distributed in the principle salivary gland, but not in the accessory salivary gland or other tissues (Fig. 1e). For most piercing-sucking insects, two types of saliva (gel and watery saliva) are secreted into plant tissues. The salivary sheath, formed from gel saliva, serves as an insoluble lining along the stylet path, potentially providing a scaffold for effector delivery [29]. The salivary sheath secreted by *B. tabaci* was also stained with the anti-BtRDP serum. The BtRDP signal was mainly detected in the protuberant structure of sheath (Fig. 1f), suggesting that BtRDP attaches to salivary sheath after secretion.

BtRDP is important for *B. tabaci* feeding on tobacco plants

RNA interference was conducted to investigate the role of BtRDP in whitefly feeding. Treatment with dsBtRDP efficiently and specifically suppressed the target gene at both the transcript and protein levels (Fig. 2a; Fig. S5). There was no significant difference in the salivary sheath morphology between dsBtRDP-treated and dsGFP-treated *B. tabaci* (Fig. S6a). However, the salivary sheath length of dsBtRDP-treated *B. tabaci* was significantly shorter than that of the dsGFP-treated control (Fig. S6b). Furthermore, treatment with dsBtRDP did not affect the survivorship of whiteflies (Fig. S6c). A reproduction analysis was conducted, which indicated that the dsBtRDP-treated whiteflies oviposited fewer eggs than the control (Fig. 2b). The feeding behavior of whiteflies was monitored using the electrical penetration graph (EPG) technique, which distinguishes nonpenetration (np), pathway (C), phloem salivation (E1), and phloem ingestion (E2) phases [30] (Fig. S7). Compared to the dsGFP control, dsBtRDP-treated *B. tabaci* exhibited a significant reduction in phloem ingestion and an extended pathway duration, indicating that BtRDP is necessary for efficient feeding (Fig. 2c).

Salivary proteins are reported to function in intercellular space and/or extracellular space after being secreted into host plants [29, 31]. To investigate the role of BtRDP in different subcellular locations of host plants, we constructed two transgenic *N. tabacum* lines overexpressing BtRDP: one carrying the full-length coding sequence that included the signal peptide (oeBtRDP), which is expected to be secreted into the apoplast (extracellular space), and the other one lacking the signal peptide (oeBtRDP^{-sp}), which is likely to be retained in the cytoplasm. Transgenic plants containing the empty vector (EV), which exerted no significant influence on insect performance compared with the WT (Fig. S8), were used as negative controls. Fecundity and two-choice assays demonstrated that BtRDP, whether localized in the apoplast (oeBtRDP) or cytoplasm (oeBtRDP^{-sp}), enhanced whitefly settling and oviposition relative to the EV controls (Fig. 2d–i; Fig. S9), indicating that BtRDP promotes whitefly feeding behavior irrespective of its subcellular location.

BtRDP interacts with the NtRLP4/NtSORBIR1 complex

To investigate the underlying mechanism of BtRDP in improving whitefly feeding, yeast two-hybrid (Y2H) screening against a *N. benthamiana* cDNA library was conducted using BtRDP^{-sp} as a bait. Totally seven proteins were found to potentially interact with BtRDP^{-sp} (Table S2), including an NbRLP4 (accession Niben261Chr07g1310001.1 in Niben261 genome). NbRLP4 possesses a predicted N-terminal signal peptide, a malectin-like domain, a LRR domain, and a transmembrane domain. NbRLP4 and its closest homology in *N. tabacum* (NtRLP4; GenBank accession: XM_016601109) did not contain the kinase domain (Fig. 3a).

The relative expression of *NtRLP4* in response to *B. tabaci* infestation was investigated. Based on qRT-PCR results, the transcript level of *NtRLP4* was not significantly influenced by *B. tabaci* (Fig. S10). However, a decrease in the protein level of NtRLP4 was detected by Western-blotting assay

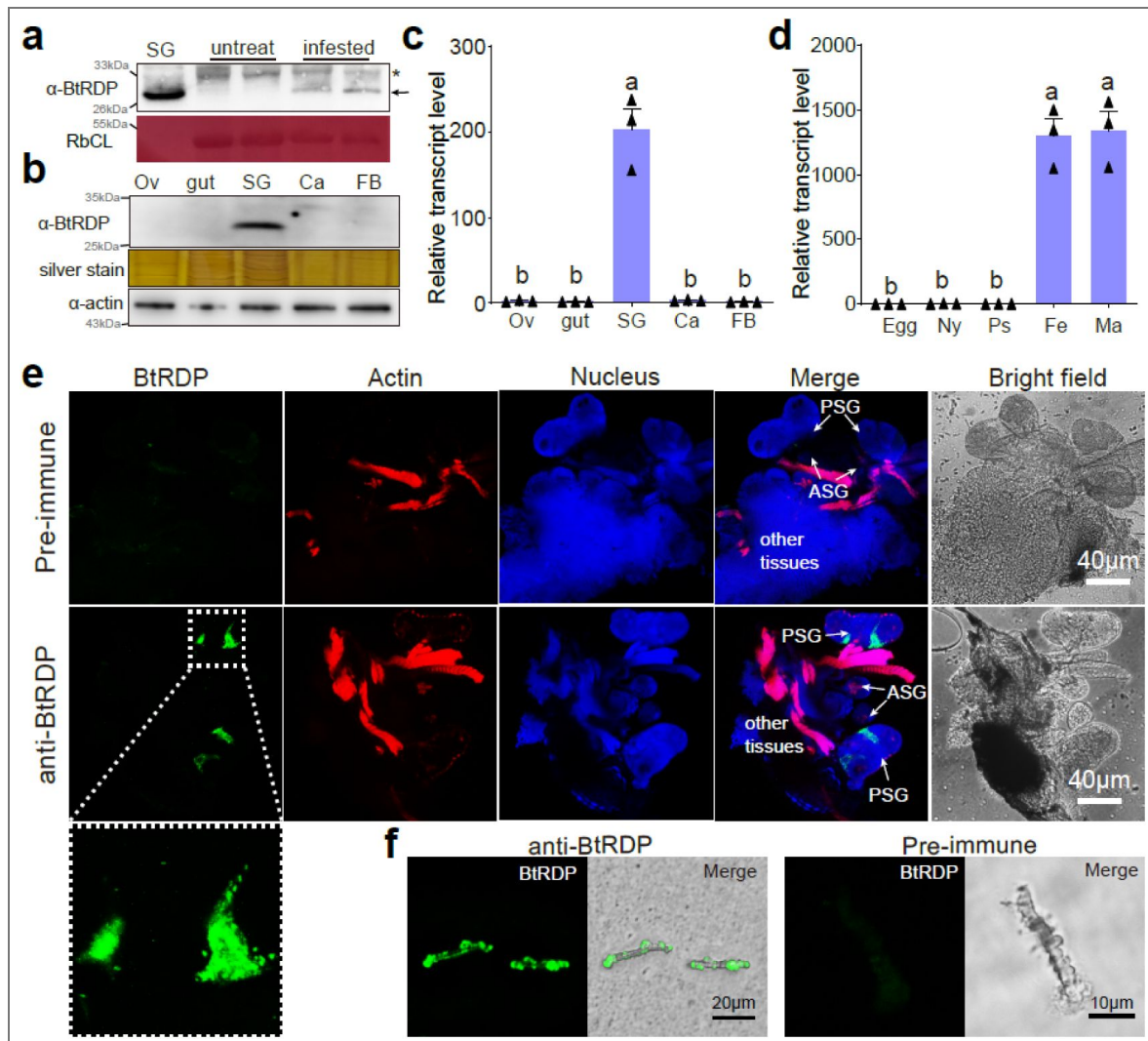


Fig. 1. BtRDP is a salivary protein and secreted into plants.

(a) Detection of BtRDP in *Nicotiana tabacum* plants. The untreated and *Bemisia tabaci*-infested tobacco plants, as well as the salivary gland samples, are collected for western-blotting assays. Asterisk indicates non-specific binding. Rubisco staining (RbCL) is conducted to visualize the amount of sample loading. (b, c) Detection of BtRDP in different tissues by western-blotting and qRT-PCR assays. Ov, ovary; SG, salivary gland; Ca, carcass; FB, fat body. Silver staining and anti-actin serum are used to visualize the sample loading. (d) Expression patterns of BtRDP in different development stages. Ny, nymph; Ps, pseudopupa; Fe, female; Ma, male. *B. tabaci tubulin* is used as an internal control. The relative quantitative method ($2^{-\Delta\Delta C_t}$) is used to evaluate the quantitative variation. Data are presented as mean values $\pm$ SEM ($n = 3$ independent biological replicates). Different lowercase letters indicate statistically significant differences at the $P < 0.05$ level according to one-way ANOVA test followed by Tukey's multiple comparisons test. (e, f) Immunohistochemical staining of BtRDP in salivary glands (e) and salivary sheath secreted from *B. tabaci* (f). The salivary gland and its nearby tissues are dissected and incubate with anti-BtRDP serum or pre-immune serum conjugated with Alexa Fluor™ 488 NHS Ester (green) and actin dye phalloidinrhodamine (red). The nucleus is stained with DAPI (blue). PSG, principal salivary gland; ASG, accessory salivary gland. The lower images represent the enlarged images of the boxed area in the upper image. Experiments are repeated thrice for (a and b), while twice for (e and f). Similar results are observed and representative images are displayed.

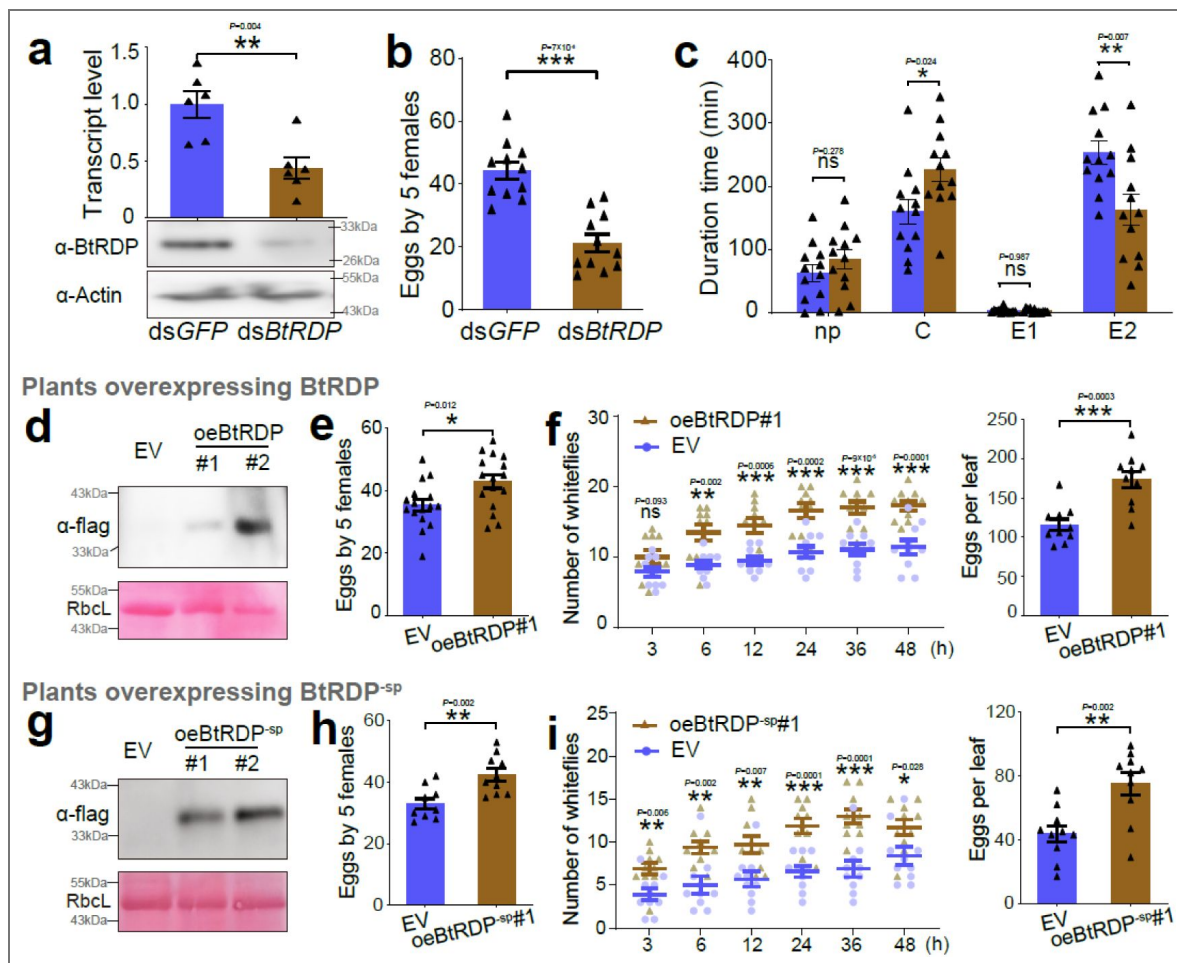


Fig. 2. Effects of BtRDP on *Bemisia tabaci*.

(a) Treating *B. tabaci* with dsBtRDP significantly reduces the transcript and protein level of target gene. The dsGFP-treated *B. tabaci* is used as a control. (b–c) Effects of BtRDP knockdown on insect reproduction (b) and feeding behavior (c). Electrical penetration graph (EPG) is used to monitor the insect feeding behavior, which can be classified into nonpenetration (np), pathway duration (C), phloem salivation (E1) and phloem ingestion (E2) phases. All EPG recordings are performed for 8 h. Typical EPG waveforms are displayed in Fig. S7. (d–f) Insect performance on transgenic plants overexpressing BtRDP (oeBtRDP). (d) Detection of BtRDP in transgenic *Nicotiana tabacum* overexpressing a complete coding region of BtRDP. The empty vector (EV) plants are used as the control. The flag tag is fused to the C-terminal ends of recombinant proteins. (e) Comparison of insect reproduction on oeBtRDP#1 and EV plants. Five *B. tabaci* individuals are confined to indicated plants for 3 days, and the oviposited eggs are counted. (f) Attraction of oeBtRDP#1 and EV leaves to *B. tabaci* in a two-choice equipment. A group of 40 female *B. tabaci* are released into a device containing oeBtRDP#1 and EV leaves. The number of insects settling on each leaf is counted. After 48 h, the number of eggs on each leaf is counted. (g–i) Insect performance on transgenic plants overexpressing BtRDP^{-sp} (oeBtRDP^{-sp}). (g) Detection of BtRDP^{-sp} in transgenic *N. tabacum* overexpressing BtRDP without a signal peptide. The insect reproduction (h) and settlement (i) on oeBtRDP^{-sp} transgenic plants are recorded. The EPG data are first checked for normality and homogeneity of variance, and data not fitting a normal distribution are subjected to log10 transformation. Data are presented as mean ± SEM. Statistical significance in panels (a), (b), (c), (e), (f), (h), and (i) was assessed using two-tailed unpaired Student's *t*-tests. ****P* < 0.001; ***P* < 0.01; **P* < 0.05; ns, not significant. Western-blotting assays are repeated thrice with the similar results.

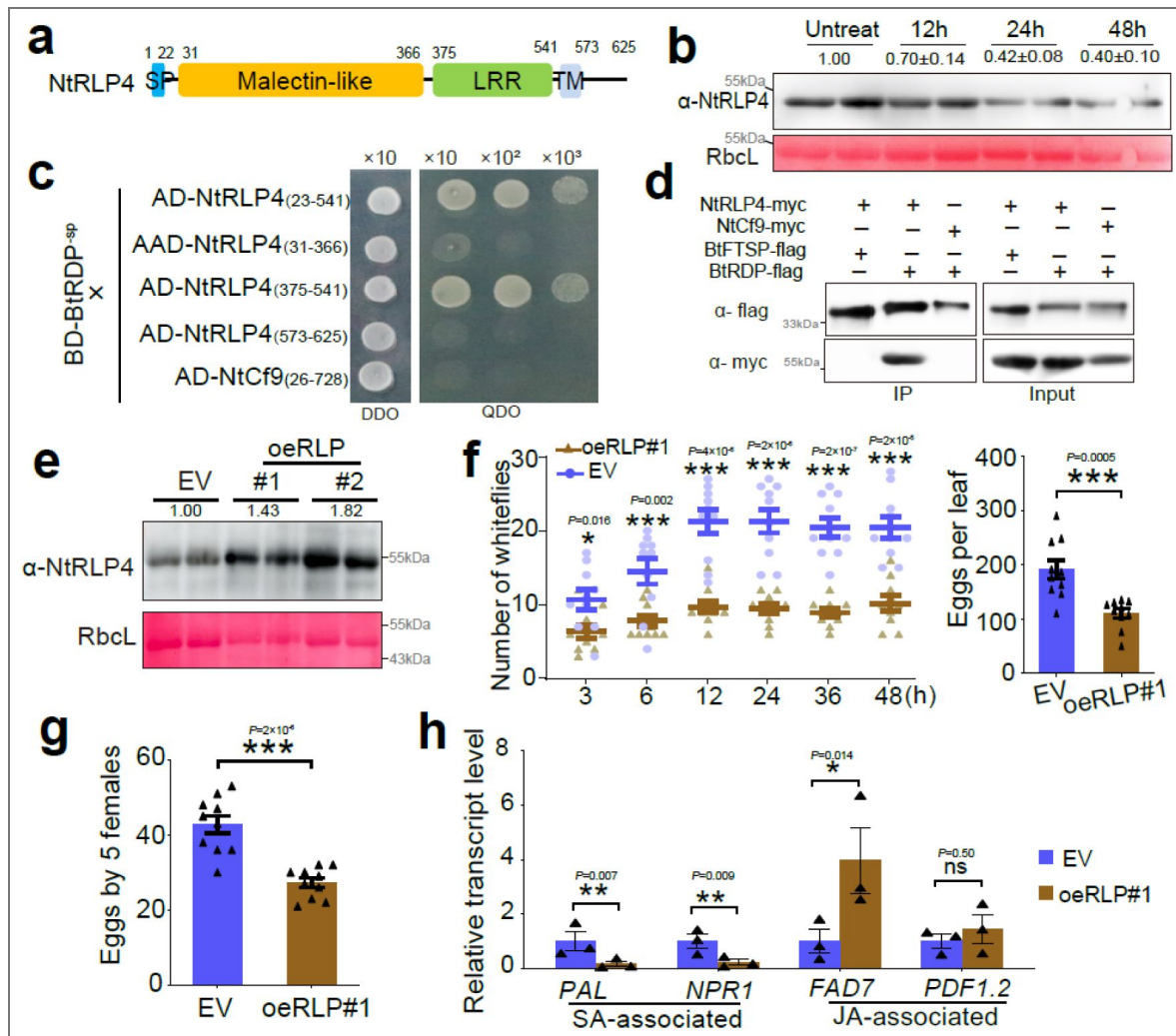


Fig. 3. NtRPL4 interacts with BtRDP and confers plant resistance to *Bemisia tabaci*.

(a) Domain organization of NtRPL4. NtRPL4 contains a predicted N-terminal signal peptide (SP), a malectin-like domain, a LRR domain, and a transmembrane domain (TM). (b) The protein level of NtRPL4 in response to *B. tabaci* infestation. The untreated *Nicotiana tabacum* is used as a negative control. Four independent biological replicates are performed, and the representative images are displayed. The band density is measured using ImageJ. The density values from 4 biological replicates are calculated and the mean value in the controls is set at 1.0. (c) Yeast two hybrid assays showing the interaction between BtRDP and NtRPL4. The different combinations of constructs are transformed into yeast cells, and are grown on the selective medium SD/-Trp/-Leu (DDO), and the interactions are tested with SD/-Trp/-Leu/-His/-Ade (QDO). (d) Co-immunoprecipitation assay showing the interaction between BtRDP and NtRPL4. Total proteins are extracted from *N. benthamiana* leaves transiently co-expressing NtRPL4-myc/NtCf9-myc with BtFTSP-flag/BtRDP-flag. All genes are expressed with a complete coding region, and the myc/flag tags are fused at the C-terminus. Precipitation is performed using flag beads. The samples are probed with anti-flag and anti-myc antibodies for immunoblotting analysis. (e-h) Analysis of transgenic *N. tabacum* overexpressing NtRPL4 (oeRPL). (e) Detection of NtRPL4 level in oeRPL plants. The empty vector (EV) plant is used as a control. Two independent oeRPL lines are selected. The samples are probed with an anti-NtRPL4 antibody. Rubisco staining (RbcL) is used to visualize the amount of sample loading. (f) Attraction of oeRPL#1 and EV leaves to *B. tabaci* in a two-choice equipment. A group of 40 female *B. tabaci* are released into a device containing oeRPL#1 and EV leaves. The number of insects settling on each leaf is counted at each time point. After 48 h, the number of eggs on each leaf is counted. (g) Comparison of insect reproduction on oeRPL#1 and EV plants. Five *B. tabaci* individuals are confined to indicated plants for 3 days, and the oviposited eggs are counted. (h) Relative transcript levels of salicylic acid (SA)- and jasmonic acid (JA)-associated genes in oeRPL#1 and EV plants. PAL, phenylalanine ammonia lyase; NPR1, nonexpressor of pathogenesis-related protein 1; FAD7, fatty acid desaturase 7; PDF1.2, plant defensin 1.2. Two independent biological replicates are performed in (d) and (e). Data in (f), (g), and (h) are presented as mean values $\pm$ SEM. For insect bioassays in (f) and (g), $n = 10$ independent biological replicates. For qRT-PCR in (h), $n = 3$ independent biological replicates. P -values are determined by two-tailed unpaired Student's t test. *** $P < 0.001$; ** $P < 0.01$; * $P < 0.05$; ns, not significant.

(Fig. 3b [↗](#)). Pairwise Y2H assays revealed that NtRLP4₍₂₃₋₅₄₁₎ (a truncated version lacking the signal peptide and transmembrane domains) interacted with BtRDP^{-SP} (Fig. 3c [↗](#)). To investigate the specificity of the NtRLP4-BtRDP interaction, another RLP protein NtCf9 with the highest sequence similarity to tomato resistance protein Cf9 (Fig. S11) [\[32\]](#), as well as three randomly selected salivary proteins from *B. tabaci* (BtFTSP, QHB15613; BtSP16.3, MN738005; BtSP37.4, MN738158), were used as controls. As a result, BtRDP^{-SP} did not interact with the truncated NtCf9 construct, in which the signal peptide and transmembrane domains were removed (Fig. 3c [↗](#)). Also, NtRLP4₍₂₃₋₅₄₁₎ could not interact with the other three salivary proteins without signal peptides (Fig. S12a, b). These findings raise the question of which domain of NtRLP4 is responsible for binding to BtRDP, as identifying the interacting domain can help to infer where the salivary protein contacts the receptor in plants. We therefore dissected the NtRLP4 domains accordingly. The results showed that it was the LRR domain, rather than the malectin-like domain or the short intercellular region, that was required for the NtRLP4-BtRDP interaction (Fig. 3c [↗](#)). To investigate the shortest LRR sequence responsible for BtRDP interaction, the LRR domain was divided into three segments. However, none of these segments interacted with BtRDP^{-SP} (Fig. S12a). Co-immunoprecipitation (Co-IP) assays further confirmed the specific interaction between BtRDP and NtRLP4 (Fig. 3d [↗](#)). It was the C-terminal myc tag-fused NtRLP4 (NtRLP4-myc), but not the NtCf9 (NtCf9-myc), that could be immunoprecipitated by the C-terminal flag tag fused BtRDP (BtRDP-flag). In the bimolecular fluorescence complementation (BiFC) assay, YFP fluorescence was observed upon co-expression of the N-terminal nYFP tag fused NtRLP4 and N-terminal cYFP tag fused BtRDP^{-SP} (Fig. S12c). Furthermore, in oeBtRDP transgenic plants, endogenous NtRLP4 was specifically immunoprecipitated with BtRDP-flag, which was significantly different from the control (Fig. S12d). These results suggest that BtRDP interacts with NtRLP4.

The absence of kinase motifs in the C-terminal region of NtRLP4 suggests that it may function in the cooperation with other proteins to initiate signaling. The receptor-like kinase SOBIR1, which contained a kinase domain (Fig. S13a), has been widely reported to be required for the function of LRR-RLPs in the innate immunity [\[6\]](#). However, whether SOBIR1 interacted with malectin-LRR RLP remains largely unknown. Through Co-IP assays, our study demonstrated that NtSOBIR1 was capable of interacting with NtRLP4 but not BtRDP (Fig. S13b, c).

NtRLP4 and NtSOBIR1 confer resistance to *B. tabaci* infestation

The transgenic *N. tabacum* overexpressing NtRLP4 (oeRLP#1 and oeRLP#2, without any tags) was constructed (Fig. 3e [↗](#)). Compared with the EV control, both oeRLP lines were less attractive to *B. tabaci*, alongside less settling and ovipositing insects (Fig. 3f [↗](#); Fig. S14a). Also, *B. tabaci* had a significantly decreased fecundity in oeRLP plants (Fig. 3g [↗](#); Fig. S14b). Similar results were observed after transient overexpression of NtRLP4-GFP in *N. tabacum* (Fig. S14c, d), indicating that NtRLP4 is resistant to *B. tabaci* infestation.

Thereafter, the EV and oeRLP#1 plants underwent transcriptomic sequencing (Fig. S15a). A total of 729 differentially expressed genes (DEGs) were identified, comprising 423 up-regulated and 306 down-regulated genes (Table S3). Enrichment analysis demonstrated that the majority of up-regulated DEGs were associated with plant-pathogen interaction, environmental adaptation, MAPK signaling pathway, and signal transduction (Fig. S15b). In contrast, the glutathione metabolism, carbohydrate metabolism, and amino acid metabolism pathways were significantly enriched in down-regulated DEGs (Fig. S15c). Noteworthy, numerous DEGs were annotated as RLK/RLP or WRKY transcription factor, with the majority of them being significantly upregulated (Fig. S15d, e). These results suggest an enhanced defense response in oeRLP plants. The crosstalk between salicylic acid (SA) and jasmonic acid (JA) pathways is critical for plant defenses [\[14\]](#). The increased defense of oeRLP transgenic plants was partially caused by the altered hormonal signaling. The JA-associated gene *FAD7* was significantly induced, while the SA-associated genes *PAL* and *NPR1* were repressed in oeRLP plants (Fig. 3h [↗](#)). Transient overexpression of NtRLP4-GFP in *N. tabacum* plants via *Agrobacterium* infiltration exerted similar effects (Fig. S14e). Furthermore, NtRLP4-GFP significantly enhanced H₂O₂ accumulation (Fig. S14f).

The function of NtSOBIR1 was also investigated by transient overexpression and hairpin RNAi (Fig. S16). Overexpression of NtSOBIR1-GFP induced the cell death phenotype five days post agro-injection (Fig. S16). Insects preferred to settle and oviposit on GFP-expressed *N. tabacum* than on NtSOBIR1-GFP-expressed control, even though no obvious death phenotype was observed two days post agro-injection (Fig. S16c, d). For NtSOBIR1 silencing, no cell death phenotype was observed, while the NtSOBIR1-silenced leaves exhibited reduced growth (Fig. S16f). Compared with the EV-silenced control, NtSOBIR1-silenced *N. tabacum* was slightly attractive to *B. tabaci* (Fig. S16g, h). Collectively, these results demonstrate that NtSOBIR1 confers resistance to *B. tabaci* infestation.

BtRDP attenuates plant defenses by promoting the degradation of host RLP4

To investigate the interplay between NtRLP4 and BtRDP, transgenic *N. tabacum* silencing NtRLP4 (*RNAi-RLP#1* and *RNAi-RLP#2*) plants were constructed. Both *RNAi-RLP* lines showed reduced NtRLP4 levels compared with EV plants, with *RNAi-RLP#2* exhibiting a stronger silencing effect (Fig. S17a). Compared with EV plants, *B. tabaci* showed a tendency to settle on NtRLP4-silenced plants and lay more eggs (Fig. S17b, c). Thereafter, ds*GFP*- and ds*BtRDP*-treated *B. tabaci* were reared on EV, *RNAi-RLP#1*, and *RNAi-RLP#2*, respectively. The ds*BtRDP*-treated *B. tabaci* produced less offspring than the control (decreased by 28.3%; $P=0.0007$) on EV plants (Fig. 4a), consistent with the results obtained from WT plants (Fig. 2b). On the *RNAi-RLP#1* plant, the impaired insect reproduction was still observed after ds*BtRDP* treatment, but the extent was slightly reduced (decreased by 23.7%; $P=0.002$) (Fig. 4a). On the *RNAi-RLP#2* plant, there was no significant difference between ds*GFP* and ds*BtRDP* treatments (Fig. 4a). The differential rescue effect between the two RNAi lines might result from their different NtRLP4 silencing efficiencies, with the lower NtRLP4 level in *RNAi-RLP#2* leading to the more complete rescue phenotype (Fig. S17a; Fig. 4a). Collectively, these results suggest that *RNAi-RLP* plants are beneficial for insects and can partially, but not completely, rescue the impaired feeding performance of ds*BtRDP*-treated insects.

To investigate the influence of BtRDP on NtRLP4, GFP-tagged NtRLP4 (NtRLP4-GFP) was transiently co-expressed with mCherry-tagged BtRDP (BtRDP-mCherry), red fluorescent protein (RFP)-mCherry, and BtFTSP1-mCherry through agroinfiltration, respectively. All tags were fused with the C-terminal region. As discovered, BtRDP-mCherry did not influence the location patterns of NtRLP4-GFP (Fig. S18). However, a significant reduction in the fluorescent signal of NtRLP4-GFP was detected after co-expression with BtRDP-mCherry, but not with the controls (Fig. S18). Besides, co-expression with BtRDP-mCherry/NtRLP4-GFP significantly reduced H_2O_2 accumulation (Fig. 4b), suppressed the expression of JA-associated genes *FAD7* and *PDF.12*, and induced that of the SA-associated genes *PAL* and *NPR1* (Fig. 4c). It was discovered by insect bioassays that, transient overexpression of BtRDP-mCherry rescued the negative effects on whitefly performance caused by NtRLP4-GFP overexpression (Fig. 4d, e). In the host choice experiments, *B. tabaci* displayed a preference for settling on *N. tabacum* leaves co-expressing BtRDP-mCherry/NtRLP4-GFP over those co-expressing RFP-mCherry/NtRLP4-GFP (Fig. 4d). However, almost no preference was observed in *B. tabaci* feeding on *N. tabacum* plants co-expressing BtRDP-mCherry/NtRLP4-GFP and plants expressing GFP alone (Fig. S19).

The attenuated NtRLP4-GFP fluorescent signal (Fig. S18), along with the observed decrease in the NtRLP4 protein level during *B. tabaci* infestation (Fig. 3b), led us to speculate on whether BtRDP influenced the accumulation of NtRLP4. To this end, the BtRDP^{-SP}-his and GFP-his proteins were prokaryotically expressed in *Escherichia coli*. Different concentrations of purified BtRDP^{-SP}-his and GFP-his proteins were infiltrated into *N. tabacum* plants overexpressing NtRLP4-myc. Interestingly, the protein level of NtRLP4-myc slightly decreased with the increasing amount of BtRDP^{-SP}-his, while no such effect was observed with GFP-his (Fig. S20). Moreover, the BtRDP^{-SP}-his protein did not have any significant effect on the accumulation of another RLP NtCf9-like (Fig.

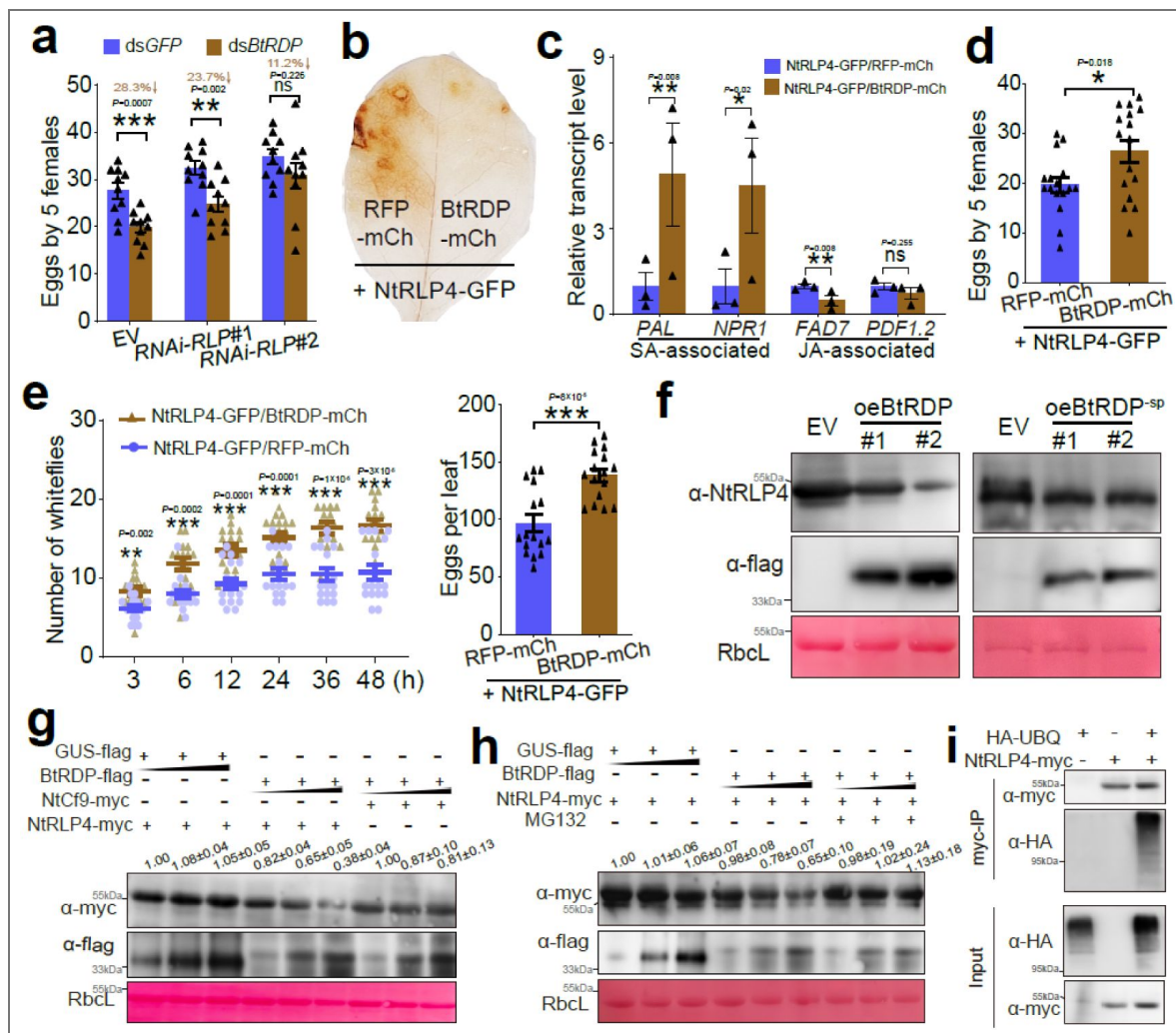


Fig. 4. BtRDP suppress plant defenses by promoting NtRLP4 degradation.

(a) Effects of dsBtRDP suppression on insect reproduction ($n = 10$ independent biological replicates) when feeding on empty vector (EV) and NtRLP4-silenced (*RNAi-RLP*) transgenic *Nicotiana tabacum*. (b) Transient overexpressing BtRDP-mCherry attenuates H_2O_2 accumulation caused by NtRLP4-GFP overexpression. *N. tabacum* co-expressing RFP-mCherry and NtRLP4-GFP is used as a control. All genes are expressed with a complete coding region. The experiment is repeated 5 times with the similar results. (c) Relative transcript level of salicylic acid (SA)- and jasmonic acid (JA)-associated genes in BtRDP-mCherry/NtRLP4-GFP and RFP-mCherry/NtRLP4-GFP plants ($n = 3$ independent biological replicates). PAL, phenylalanine ammonia lyase; NPR1, nonexpressor of pathogenesis-related protein 1; FAD7, fatty acid desaturase 7; PDF1.2, plant defensin 1.2. (d) Comparison of insect reproduction ($n = 16$ independent biological replicates) on transient-expressed BtRDP-mCherry/NtRLP4-GFP and RFP-mCherry/NtRLP4-GFP plants. (e) Attraction of BtRDP-mCherry/NtRLP4-GFP and RFP-mCherry/NtRLP4-GFP leaves to *B. tabaci* in a two-choice equipment ($n = 16$ independent biological replicates). The number of eggs on each leaf is counted at 48 h post insect release. (f) Effects of BtRDP and BtRDP^{sp} overexpression on NtRLP4 accumulation. Transgenic oeBtRDP (left) and oeBtRDP^{sp} (right) plants are probed with anti-NtRLP4 and anti-flag antibodies for immunoblotting analysis. The experiments are repeated twice with the similar results. (g) Degradation of NtRLP4 by BtRDP in *N. benthamiana* leaves. NtRLP4-myc and BtRDP-flag are transiently co-expressed in *N. benthamiana* plants through *Agrobacterium* infiltration. *Agrobacterium* carrying NtCf9-myc and GUS-flag are used as negative controls. (h) Effects of 26S proteasome inhibitor (MG132) on NtRLP4 accumulation. Co-infiltrated leaves are treated with MG132 at 24 h post injection. The samples are probed with anti-flag and anti-myc antibodies for immunoblot analysis. Rubisco staining (Rbcl) is conducted to visualize the amount of sample loading. The small triangle indicates the different concentrations of *Agrobacterium* ($OD_{600} = 0.05, 0.3, \text{ and } 1.0$). (i) NtRLP4 is ubiquitinated *in planta*. NtRLP4-myc is co-expressed transiently with HA-UBQ in *N. benthamiana* leaves. Extracted total proteins are immunoprecipitated by anti-myc beads and immunoblotted with anti-myc or anti-HA antibody. Experiments are repeated thrice for (g) and (h), while twice for (i). Band density is measured using ImageJ. The density values from three biological replicates are calculated and the mean value in the first lane is set at 1.0. Data are presented as mean values $\pm$ SEM. P -values are determined by two-tailed unpaired Student's t test. *** $P < 0.001$; ** $P < 0.01$; * $P < 0.05$; ns, not significant.

S20). We further assessed the impact of BtRDP on NtSOBIR1 following NtRPL4 destabilization. As the BtRDP-flag amount increased, NtRPL4-myc accumulation was markedly reduced, while NtSOBIR1-flag levels remained unchanged (Fig. S21).

Then, the NtRPL4 levels in oeBtRDP and oeBtRDP^{-sp} plants were detected. Compared with the EV control, a reduced NtRPL4 abundance was observed in both oeBtRDP#1 and oeBtRDP#2 transgenic lines, but not in the oeBtRDP^{-sp} plants (Fig. 4f), suggesting the importance of BtRDP secretion in extracellular space. BtRDP attached to salivary sheath localized in plant apoplast (Fig. 1f). The presence of signal peptide promoted the secretion of BtRDP into the extracellular space. Therefore, the complete coding region of *BtRDP* was used in subsequent experiments. The NtRPL4-myc and BtRDP-flag were transiently co-expressed in *N. benthamiana* plants through *Agrobacterium* infiltration. As a result, BtRDP-flag significantly decreased the protein level of NtRPL4-myc, but not NtCf9-myc (Fig. 4g). qRT-PCR analyses demonstrated that the transcript level of *NtRPL4* remained unchanged (Fig. S22). These findings indicate that the reduction in the protein level of NtRPL4 is not caused by transcriptional changes but rather by degradation.

The ubiquitin system and autophagy are two major pathways regulating protein degradation [33]. At first, MG132, a 26S proteasome inhibitor known to inhibit the degradation of ubiquitin-conjugated proteins, was employed [34]. The results showed that MG132 efficiently prevented the degradation of NtRPL4-myc by BtRDP-flag (Fig. 4h). Then, the autophagy inhibitor 3-Methyladenine (3-MA) and bafilomycin A1 (BAF) that can block autophagy in plants were used [35]. However, the decreased protein level of NtRPL4-myc was still observed in the presence of 3-MA or BAF (Fig. S23). Additionally, NtRPL4-myc and HA-tagged ubiquitin (HA-UBQ) were co-expressed in *N. benthamiana*. Total proteins were extracted from *N. benthamiana* leaves and then subjected to immunoprecipitation with anti-myc beads. The ubiquitin attached to NtRPL4 was determined by immunoblot using the anti-HA antibody. Leaves expressing NtRPL4-myc or UBQ-HA alone were used as negative controls. The results showed that a strong UBQ signal was detected in the immunoprecipitated product only when NtRPL4-myc and UBQ-HA were co-expressed (Fig. 4i). The molecular weight of UBQ-attached NtRPL4-myc was remarkably larger than that of NtRPL4-myc itself (Fig. 4i). These results suggest that BtRDP promotes NtRPL4 degradation through the ubiquitin system, but not the autophagy pathway.

Degrading plant RLP4s by salivary effectors is a common strategy for insects

To investigate the ability of BtRDP to degrade RLPs from other host plants of *B. tabaci*, *Solanum lycopersicum* RLP4 (SlRLP4; GenBank accession: XP_004232910) was selected for further analysis. Sequence analysis revealed a similar gene structure between SlRLP4 and NtRPL4 (Fig. S24; Fig. S25a). Moreover, the interaction between BtRDP and SlRLP4 was confirmed through Y2H and Co-IP assays (Fig. S25b, c). Additionally, SlRLP4-myc and SlCf9-like-myc were transiently co-expressed with varying concentrations of BtRDP-flag or GUS-flag, with the myc and flag tags fused with the C-terminus. The results showed that BtRDP-flag exerted no influence on the accumulation of SlCf9-like-myc protein. In contrast, the protein level of SlRLP4-myc was significantly reduced in the presence of BtRDP-flag (Fig. S25d), indicating the conserved function of BtRDP in *B. tabaci* adaptation to different host plants.

BtRDP was a gene specific to the Aleyrodidae family, lacking homologues in other species. To explore whether the degradation of RLPs by salivary effectors was a conserved mechanism among different herbivorous insects, a RLP4 homology from *Oryza sativa* (OsRLP4; GenBank accession: XP_015645303; Fig. 5a) and twenty salivary proteins from planthopper species lacking signal peptides were paired to identify the potential interactions using Y2H assays. As a result, only the salivary protein NISP104^{-sp} (*N. lugens* salivary protein consisting of 104 amino acids, GenBank accession: MF278694) interacted with OsRLP4₍₂₉₋₅₅₁₎ (OsRLP4 lacking signal peptides and transmembrane domains), while the other salivary proteins tested did not show any interaction (Fig. 5b, c; Fig. S26). NISP104 did not exhibit any sequence or structural similarity to BtRDP (Fig. S27). Truncation analysis revealed that NISP104^{-sp} interacted with both the LRR domain and the malectin-like domain of OsRLP4 (Fig. 5c), which was slightly different from the BtRDP-NtRPL4

interaction (Fig. 3c). Given that both BtRDP and NISP104 interacted with the LRR domain of RLP4, further investigation focused on the specific region of the LRR domain responsible for NISP104 binding. The LRR domain of OsRLP4 was divided into three segments, mirroring the division of the LRR domain in NtRLP4 (Fig. S12a). NISP104^{-SP} interacted with the truncated LRR domain (OsRLP4₍₄₈₀₋₅₂₆₎) (Fig. 5c). In contrast, none of the truncated LRR domain of NtRLP4 interacted with BtRDP^{-SP} (Fig. S12a). Collectively, these findings suggest that the binding sites for salivary proteins to RLP4 may differ between *N. lugens* and *B. tabaci*.

NISP104 was a planthopper-specific salivary protein nearly exclusively expressed in salivary glands (Table S1; Fig. 5d; Fig. S28). The function of NISP104 or its gene homologs in other planthopper species has not been well investigated in previous studies. Silencing NISP104 did not affect insect survivorship (Fig. 5e). However, significant reductions in honeydew excretion and impaired fecundity were observed in dsNISP104-treated *N. lugens* (Fig. 5f, g), indicating the importance of NISP104 in insect feeding.

Subsequently, OsRLP4-myc was transiently co-expressed with a complete coding region of NISP104 (C-terminal flag tagged, NISP104-flag). Another salivary protein NISP7-flag and NtCf9-myc were used as negative controls. It was found that NISP104-flag specifically and significantly decreased the protein level of OsRLP4-myc, which differed significantly from that of the NISP7-flag (Fig. 5h). Additionally, NISP104-flag failed to decrease the protein level of NtCf9-myc (Fig. 5h). These findings suggest that *N. lugens* potentially evolves salivary effectors to target rice RLPs.

Discussion

Plants have evolved a complex innate immune system to protect themselves against pathogen/insect invasion, and the cell surface-localized PRRs function on the frontline [4]. To counteract this response, the pathogens have developed various strategies to avoid plant PRR-mediated PTI for their own advantage [36]. However, it remains unknown whether insects take the similar strategy to sabotage plant immunity. In this study, we demonstrate that the RLP4 confer plant resistance against herbivorous insects. In response, both planthoppers and whiteflies secrete salivary effectors to facilitate the degradation of defensive RLP4, leading to the attenuated plant immunity. Our results suggest that manipulation of PRRs may be a conserved strategy for herbivorous insects.

PRRs detect modified-self or non-self patterns and initiate a cascade of immune signaling events, which subsequently regulate downstream defense responses, conferring plant resistance against invaders [4, 5]. Although various insect salivary components have been identified as triggers for plant immune responses, the specific PRRs involved in this process remain unclear. Our study indicates that RLP4 significantly influences insect performance by modulating plant hormones and ROS (Fig. 3f-h; Fig. S14), highlighting the critical role of RLP4 in plant immunity. However, the precise ligand for RLP4 has yet to be identified. The malectin-LRR receptor RLP4 may respond to the DAMPs or HAMPs released during insect feeding [37], a hypothesis that warrants further investigation. RLP4 lacks an intracellular kinase domain, and instead forms bimolecular receptor kinases by interacting with SOBIR1 (Fig. S13). Silencing of NtRLP4 or NtSOBIR1 in *N. tabacum* attracts *B. tabaci* and promotes insect reproduction, whereas overexpression of either gene exerts the opposite effect (Fig. S14-S17). The association between RLPs and SOBIR1 protein kinases is structurally and functionally analogous to that of genuine receptor kinases [6]. Therefore, destabilizing the RLP4/SOBIR1 complex can potentially disrupt the immune recognition processes. Noteworthy, although NtRLP4 interacts with SOBIR1, this alone does not confirm that it operates strictly through this canonical module. Evidence from other RLPs shows that co-receptor usage can be flexible, and some RLPs function partly or conditionally independent of SOBIR1 [38]. Therefore, a more definitive assessment of NtRLP4 signaling will therefore require genetic dissection of its co-receptor dependencies, including but not limited to SOBIR1.

As pivotal regulators in immune signaling, PRRs are frequently targeted by multiple pathogens through diverse mechanisms. For example, *Pseudomonas syringae* hampers the functionality of receptor complexes by disrupting the ligand-dependent association between receptor-like kinase

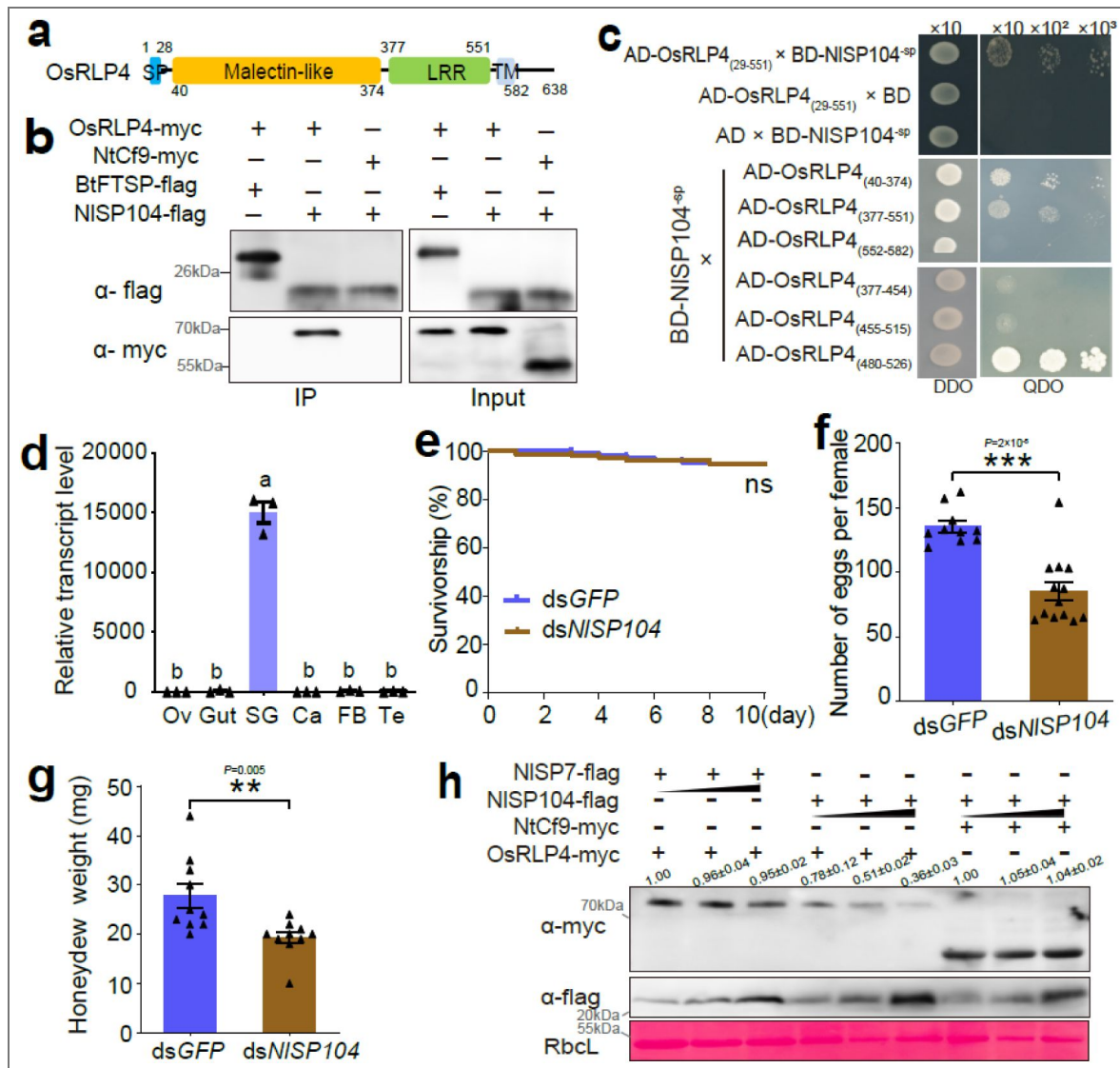


Fig. 5. Rice RLP4 are targeted by salivary protein NISP104 from *Nilaparvata lugens*.

(a) Domain organization of *Oryza sativa* RLP4 (OsRPL4). (b, c) Yeast two-hybrid and Co-IP assays showing the interaction between OsRPL4 and NISP104. All genes are expressed with a complete coding region in Co-IP assays, while NISP104 without a signal peptide is used in yeast two-hybrid assays. Experiments in (b) are repeated twice with the similar result. (d) The expression patterns of *NISP104* in different *N. lugens* tissues. Te, testis; Ov, ovary; SG, salivary gland; Ca, carcass; FB, fat body. Data are presented as mean values $\pm$ SEM ($n = 3$ independent biological replicates). Different lowercase letters indicate statistically significant differences at $P < 0.05$ level according to one-way ANOVA test followed by Tukey's multiple comparisons test. (e-g) Effects of dsRNA treatment on insect survivorship (e), reproduction (f), and honeydew excretion (g). For survivorship analysis, a group of 30 *N. lugens* are reared in a cage. Three independent biological replications are performed. Differences in survivorship between the two treatments are tested by log-rank test. ns, not significant. For reproduction analysis, $n = 10$ and 13 individuals are tested in dsGFP- and dsNISP104-treatment, respectively. The sterile females are excluded from data analysis. For honeydew analysis, $n = 10$ independent biological replicates. P -values in (f) and (g) are determined by two-tailed unpaired Student's t test. *** $P < 0.001$; ** $P < 0.01$. (h) Effect of NISP104 on the accumulation of transiently expressed OsRPL4. OsRPL4-myc is agro-injected into *Nicotiana benthamiana* together with different concentrations of NISP104-flag and NISP7-flag. NtCf9-myc is used as a negative control. The small triangle indicates the different concentrations ($OD_{600} = 0.05, 0.3$, and 1.0) of *Agrobacterium*. Rubisco staining (RbCL) is conducted to visualize the amount of sample loading. Experiments are repeated three times with the similar results. Band density is measured using ImageJ. The density values from three biological replicates are calculated with the mean value in the first lane being set at 1.0.

BAK1 and the flagellin receptor FLS2 [24], while *Phytophthora infestans* promotes alternative splicing of the host RLPs to enhance infection [23]. PRRs are usually subjected to degradation via post-translational modifications [39]. In plants, the activity and stability of PRRs are under stringent regulation since excessive activation of PRRs could be detrimental [22, 40]. One prevalent regulatory mechanism involves ubiquitination followed by subsequent degradation mediated by the 26S proteasome [39]. Mounting evidence has demonstrated that bacteria, fungi, viruses, and oomycetes exploit such mechanisms to facilitate PRR degradation [41–44]. Nevertheless, the occurrence of effector-mediated PRR degradation in insects is unreported. Our study reveals that the salivary effectors BtRDP from *B. tabaci* and NISP104 from *N. lugens* interacts with plant RLP4s, resulting in their degradation in a ubiquitin-dependent manner (Fig. 4f–i [45]).

Notably, BtRDP and NISP104 exhibit neither sequence nor structural similarity and do not resemble the known eukaryotic ubiquitin-ligase domains. Their interaction with RLP4s occurs in the extracellular space (Fig. 3d [46]; Fig. 5c [47]), whereas the ubiquitin-proteasome system primarily functions within the cytosol and nucleus [45]. Furthermore, NtRLP4 reduction is observed only in oeBtRDP transgenic plants, not in oeBtRDP^{−sp} plants (Fig. 4f [48]), indicating that BtRDP exerts its influence on NtRLP4 in the extracellular space. These observations collectively refute the possibility that BtRDP or NISP104 possesses the intrinsic E3 ligase activity capable of directly ubiquitinating RLP4s within plant cells. Importantly, the reduced NtRLP4 levels may not stem from a direct physical interaction between BtRDP and NtRLP4. Instead, BtRDP may indirectly affect RLP4 post-translational modification, thereby accelerating its degradation, which warrants further investigation. Additionally, the independent evolution of RLP4-targeting effectors in various insect lineages may expedite plant-insect co-evolution. Plants with robust RLP4-mediated defenses can select for herbivores carrying effective effectors, driving an evolutionary “arms race”. Conversely, insects targeting similar host proteins may exert parallel selective pressures on unrelated plant species, potentially shaping defense diversification across plant communities. Understanding these dynamics provides insight into how molecular interactions scale up to influence herbivore adaptation, host range, and the ecological outcomes of plant-insect interactions.

Herbivorous insects employ a mix of salivary components to ensure successful feeding [29]. Previous research has indicated that *B. tabaci* suppresses plant defenses by utilizing SA-JA crosstalk, with the salivary protein Bt56 playing a crucial role in this process [14]. Bt56 activates the SA-signaling pathway by directly interacting with a KNOTTED 1-like homeobox transcription factor [14]. This study investigates the effects of another salivary effector, BtRDP, on plant hormones, and finds that it elicits similar effects to Bt56, albeit primarily impacting the signal recognition process (Fig. 3h [49]; Fig. S14e). This leads us to speculate that different salivary effectors from the same insect species may simultaneously target the same signaling pathway to ensure effective immune suppression. Our analysis reveals that BtRDP interacts with other host genes apart from NtRLP4 (Table S2), and silencing NtRLP4 cannot completely rescue the impaired feeding performance of dsBtRDP-treated insects (Fig. 4a [50]). Salivary proteins exert a crucial effect on mediating the interactions between herbivores and plants. The dynamic interplay between plants and insects may be highly complex, one salivary protein can target multiple plant pathways, and one plant pathway is influenced by multiple salivary proteins.

Together, this study reveals that suppressing PRR-mediated plant immunity may be a conserved strategy employed by herbivorous insects for successful feeding. We demonstrate that whiteflies and planthoppers have independently evolved salivary effectors that facilitate the ubiquitin-dependent degradation of defensive RLP4 in host plants, thereby dampen RLP4-mediated plant immunity (Fig. 6 [51]). Nevertheless, the precise mechanism by which RLP4 contributes to plant defense warrants further consideration. While it may function as a canonical PRR that perceives insect-derived molecular patterns, several lines of evidence point to an alternative interpretation. Structurally, RLP4 differs from classical LRR-RLP: it contains a malectin-like domain and a relatively small LRR domain, contrasting with typical LRR-RLPs that often possess large LRRs dedicated to ligand binding. Functionally, NtRLP4 overexpression lines exhibit significantly altered transcriptional profiles and dysregulated SA/JA pathways even in the absence of insect infestation, a phenotype inconsistent with canonical PRRs, which typically remain quiescent until ligand

perception. These findings point to an alternative explanation: rather than functioning as a classical PRR that recognizes insect-derived molecules, RLP4 may act as a regulatory component within plant immune signaling networks. Elucidating the precise mechanism of RLP4 in conferring plant defense against herbivorous insects will therefore be an important focus of future research.

Material and methods

Insects and plants

A colony of *B. tabaci* (cryptic species MED) was originally collected from a soybean field in Suzhou, Anhui Province, China. The insects were subsequently maintained on *N. tabacum* cv. K326 plants under controlled laboratory conditions (25 ± 1 °C, 50–70% relative humidity, 16:8 h light:dark photoperiod). For experimental use, *N. benthamiana* and *N. tabacum* plants were grown in a climate-controlled growth chamber (23 ± 1 °C, 16:8 h light: dark cycle).

Phylogenetic analysis of RDP

To identify potential RDP homologs across insect species, we performed a BLASTp search using the BtRDP protein sequence as a query against the predicted proteomes of 30 insect species. Hits were filtered using an E-value cutoff of 10^{-5} . The retrieved RDP homologs were aligned using MAFFT (v7.310) with default parameters [46]. To improve alignment quality, we removed ambiguously aligned regions using Gblocks with default settings. The optimal amino acid substitution model (JTT+G) was determined using ModelTest-NG (v0.1.6). Phylogenetic relationships were inferred using maximum likelihood analysis in RAxML (v0.9.0) with 1000 bootstrap replicates to assess branch support [47]. The final tree was visualized and annotated using iTOL (<https://itol.embl.de/>).

Preparation of anti-BtRDP and anti-NtRLP4 serum

The coding sequence of BtRDP (excluding the signal peptide) was PCR-amplified and cloned into the pET-28a vector (Novagen) to generate a C-terminal 6×His-tagged fusion protein. The recombinant plasmid was transformed into *Escherichia coli* Transetta (DE3) competent cells (TransGen Biotech, Beijing, China). Protein expression was induced with 0.5 mM isopropyl-β-D-thiogalactoside (IPTG) at 28°C for 8 h. Bacterial cells were harvested by centrifugation and lysed via ultrasonication for 30 min. The recombinant BtRDP^{SP}-His protein was purified under native conditions using Ni-NTA agarose (Qiagen, Hilden, Germany) according to the manufacturer's protocol.

Purified BtRDP^{SP}-His protein was used to immunize rabbits (Hua'an Biotechnology, Hangzhou, China) for custom polyclonal antibody production. The anti-NtRLP4 serum, prepared by immunizing rabbits with peptide GQNSDRPISTKNST, was produced via the custom service of Genscript Biotechnology Company (Nanjing, China). The anti-BtRDP serum was subsequently conjugated to Alexa Fluor™ 488 NHS Ester (#A20000, Thermo Fisher Scientific) following standard labeling protocols.

The specificity of anti-BtRDP serum was tested by comparing dsGFP- and dsBtRDP-treated *Bemisia tabaci*. The results showed that dsBtRDP significantly reduced the target gene. Similarly, the specificity of anti-NtRLP4 serum was tested by comparing RNAi-EV and RNAi-NtRLP4 plants, which showed a reduced NtRLP4 expression in RNAi-NtRLP4 plants.

Immunohistochemistry (IHC) staining

Dissected *B. tabaci* heads with attached salivary glands and Parafilm-embedded salivary sheaths were fixed in 4% paraformaldehyde (#E672002, Sango Biotechnology, Shanghai) for 30 min. Salivary sheaths adherent to Parafilm were similarly excised and fixed under identical conditions. After fixation, samples were washed three times with PBST (PBS containing 0.1% Tween-20) and blocked with 10% FBS for 2 h at room temperature. The tissues were then incubated overnight at 4°C with Alexa Fluor 488-conjugated anti-BtRDP serum (1:200 dilution), followed by co-staining

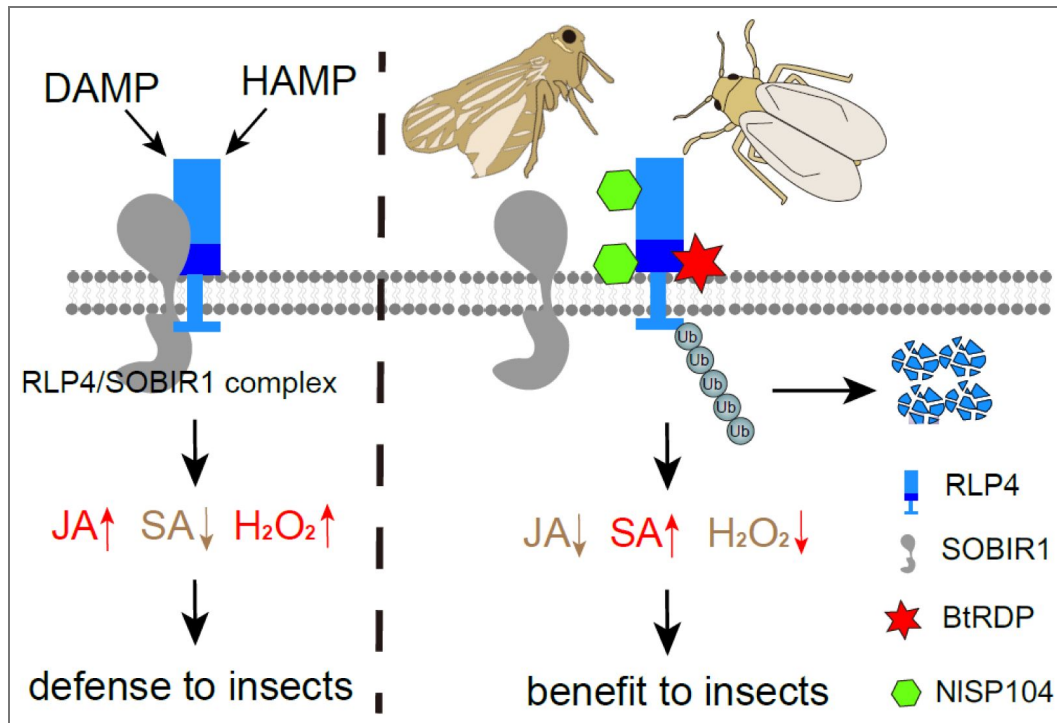


Fig. 6. The proposed model for the suppression of receptor-like proteins (RLPs)-mediated plant defenses by salivary effectors.

Host plants employ pattern-recognition receptors (PRRs) to detect various damage-associated molecular patterns (DAMPs) and herbivore-associated molecular patterns (HAMPs) triggered by insect feeding. The RLP4/SOBIR1 complex plays a vital role in initiating pattern-triggered immunity (PTI), including H_2O_2 burst, upregulation of jasmonic acid (JA), and downregulation of salicylic acid (SA), which hinders insect feeding. The whitefly *Bemisia tabaci* and planthopper *Nilaparvata lugens* independently evolved salivary proteins that targeted plant RLP4. *B. tabaci* salivary sheath protein BtRDP interacts with the leucine-rich repeat (LRR) domain of RLP4 from *Nicotiana tabaci* and *Solanum lycopersicum*, while *N. lugens* NISP104 targets both the LRR domain and the malectin-like domain of *Oryza sativa* RLP4. These interactions promote the ubiquitin-dependent degradation of RLP4, thereby disrupting the stability of the RLP4/SOBIR1 complex. The presence of salivary effectors causes a hormonal shift and suppresses the H_2O_2 burst, finally favoring insect feeding.

with phalloidin-rhodamine (1:500, Thermo Fisher Scientific) for 30 min and 4',6-diamidino-2-phenylindole (DAPI) solution (#ab104139, Abcam, Cambridge, USA) for nuclear visualization. Fluorescent images were acquired using a Leica SP8 confocal microscope (Leica Microsystems) to examine BtRDP localization patterns

Protein extraction and Western-blotting assays

To identify secreted salivary proteins in *N. tabacum* plants, approximately 200 *B. tabaci* adults were confined in a 3-cm diameter feeding cage and allowed to feed on host plants for 24 h. After feeding, the eggs deposited on the infested tobacco leaves were removed. The leaves showing no visible insect contamination were immediately frozen in liquid nitrogen and ground to a fine powder. The powdered tissue was then homogenized in RIPA Lysis Buffer (#89900, Thermo Fisher Scientific) for protein extraction. To examine the effects of insect feeding on NtRLP expression, groups of approximately 200 *B. tabaci* were allowed to feed on 4-5 true-leaf stage tobacco plants for 12, 24, or 48 h, with uninfested plants serving as controls. For silencing/overexpression efficiency tests, leaf samples were collected at 48 h post-*Agrobacterium* infiltration, along with transgenic tobacco plants at the 4-5 true-leaf stage. For insect tissue analysis, various *B. tabaci* organs were carefully dissected in 1×PBS buffer, including carcasses (20), fat bodies (20), guts (40), salivary glands (60), and ovaries (10), with sample sizes indicated in parentheses. All dissected tissues were transferred to lysis buffer for protein extraction. Total protein concentrations in both insect and plant homogenates were quantified using a BCA Protein Assay Kit (#CW0014S, CwBiotech, Taizhou, China). Equal protein amounts were separated by SDS-PAGE and electro-transferred to PVDF membranes.

Membranes were probed with the following primary antibodies: anti-FLAG (1:10,000, #MA1-91878, Thermo Fisher Scientific), anti-MYC (1:10,000, #MA1-21316, Thermo Fisher Scientific), anti-GFP (1:10,000, #MA5-15256, Thermo Fisher Scientific), anti-NtRLP4 (1:5,000, GenScript Biotechnology, Nanjing, China), and anti-BtRDP serum (1:5,000, Huaan Biotechnology, Hangzhou, China). Following primary antibody incubation, membranes were treated with HRP-conjugated secondary antibodies: goat anti-mouse IgG (1:10,000, #31430, Thermo Fisher Scientific) or goat anti-rabbit IgG (1:10,000, #31460, Thermo Fisher Scientific). Protein bands were visualized using an AI 680 image analyzer (Amersham Pharmacia Biotech, Buckinghamshire, UK), with Ponceau S staining used to verify equal protein loading.

Quantitative real-time PCR (qRT-PCR) analysis

To prepare tissue-specific samples from *B. tabaci*, adult females were dissected to isolate carcasses (10), fat bodies (10), guts (20), salivary glands (40), and ovaries (10). Additionally, different developmental stages, including eggs (50), nymphs (20), pupae (20), and adult males (20) and females (20), were collected for analysis. The number of insects in each sample is indicated in parentheses. To extract RNA from *N. tabacum* plants, the samples were grinded using liquid nitrogen after collection. Total RNA was extracted using the TRIzol Total RNA Isolation Kit (#9109, Takara, Dalian, China) following the manufacturer's instructions. RNA purity and concentration were assessed using a NanoDrop spectrophotometer (Thermo Fisher Scientific). First-strand cDNA was synthesized from equal amounts of RNA (1 µg) using the HiScript II Q RT SuperMix (#R212-01, Vazyme, Nanjing, China). Gene expression analysis was performed using a Roche LightCycler® 480 system (Roche Diagnostics, Mannheim, Germany) with SYBR Green Supermix (#11202ES08, Yeasen, Shanghai, China). The thermal cycling protocol consisted of initial denaturation at 95°C for 5 min, and amplification for 40 cycles of 95°C for 10 s and 60°C for 30 s. Primers were designed with Primer Premier v6.0 (Table S4). *B. tabaci* *Tubulin* and *N. tabacum* *Tubulin* served as internal controls for normalization. Relative gene expression was calculated using the $2^{-\Delta\Delta C_t}$ method [48]. Three to six independent biological replicates, each repeated twice, were performed.

RNA interference

The target gene DNA sequences were amplified using primers listed in Table S4 and subsequently cloned into the pClone007 Vector (#TSV-007, Tsingke Biotechnology, Beijing, China). For dsRNA production, PCR-generated DNA templates containing T7 promoter sequences were transcribed *in vitro* using the T7 High Yield RNA Transcription Kit (#TR101-01, Vazyme Biotech, Nanjing, China). The RNAi experiments were performed according to established methods [49]. Briefly, newly emerged adult whiteflies were anesthetized with carbon dioxide for 5-10 seconds. Approximately 50 nL of dsRNA solution (2 µg/µL) was microinjected into the mesothorax of each insect using a FemtoJet microinjection system (Eppendorf, Hamburg, Germany). Following injection, the whiteflies were maintained on healthy *N. tabacum* leaves under standard rearing conditions. Only actively moving insects were selected for subsequent analyses. Gene silencing efficiency was evaluated on day 4 post-injection using both qRT-PCR and Western-blotting approaches.

Generation of transgenic plants

We generated two transgenic *N. tabacum* lines expressing different BtRDP variants: one containing the full coding sequence (oeBtRDP) and another lacking the signal peptide (oeBtRDP^{SP}), which may affect protein secretion. The target sequences, along with the full-length NtRPL4 coding sequence and an NtRPL4 hairpin RNAi construct, were each cloned into the pBWA(V)HS vector. The oeBtRDP and oeBtRDP^{SP} were fused with C-terminal flag tags, while no tag was fused to oeNtRPL4. All constructs were transformed into *A. tumefaciens* strain GV3101 and used to generate transgenic tobacco plants (K326 background) through *Agrobacterium*-mediated transformation performed by Wuhan Boyuan Biological Company. Transgenic lines were verified by qRT-PCR and western blotting, with empty vector-transformed plants serving as controls, and two independent lines for each construct were selected for further experiments.

Target gene silencing in plants

The hairpin RNAi construct targeting *NtSOBIR1* was generated by cloning the specific fragment into the pBWA(V)HS plasmid using BsaI and Eco31I restriction enzymes. The recombinant vector was then transformed into *A. tumefaciens* strain GV3101 and subsequently infiltrated into *N. tabacum* leaves using the agroinfiltration method. Following a 48-h incubation period under standard growth conditions, silencing efficiency was quantitatively evaluated by qRT-PCR analysis.

Transcriptomic sequencing

To investigate the transcriptional effects of NtRPL4, BtRDP, and BtRDP^{SP} in *N. tabacum*, leaf samples from EV, oeRPL4#1, oeBtRDP#1, and oeBtRDP^{SP}#1 transgenic plants were homogenized in TRIzol Reagent (#10296018, Invitrogen, Carlsbad, CA, USA) for total RNA extraction following the manufacturer's protocol. Afterwards, total RNA was extracted according to the instructions of the manufacturer, and the RNA samples were sent to Novogene Institute (Novogene, Beijing, China) for transcriptomic sequencing. Briefly, poly(A)⁺ RNA was isolated from 20 µg pooled total RNA using oligo(dT) magnetic beads, followed by fragmentation in divalent cation buffer at 94°C for 5 min. First-strand cDNA synthesis was performed using N6 random primers, with subsequent second-strand synthesis to generate double-stranded cDNA. After end-repair and Illumina adapter ligation, the products were PCR-amplified (15 cycles) and purified using a QIAquick PCR Purification Kit (#28104, Qiagen, Hilden, Germany). The resulting cDNA libraries were quantified and quality-checked before paired-end sequencing on an Illumina NovaSeq 6000 platform. All raw sequencing data were deposited in the National Genomics Data Center under accession number PRJCA025907.

Analysis of transcriptomic data

The raw sequencing reads were quality-filtered using Illumina's internal software, and the resulting clean reads from each cDNA library were aligned to the *N. tabacum* reference genome in Sol Genomics Network

(https://solgenomics.net/ftp/genomes/Nicotiana_tabacum/edwards_et_al_2017/) using HISAT v2.1.0 [50]. Low-quality alignments were removed using SAMtools (v1.7) with default parameters [51]. Transcript abundance was quantified as Transcripts Per Million (TPM) using Cufflinks (v2.2.1) [52]. Differential expression analysis was performed with DESeq2 (v2.2.1) [53], identifying significantly DEGs using thresholds of $|\log_2(\text{fold change})| > 1$ and adjusted p -value < 0.05 . PCA of transcriptome profiles was conducted using a custom R script plotPCA (https://github.com/franco-ye/TestRepository/blob/main/PCA_by_deseq2.R). For functional annotation, KEGG pathway enrichment analysis was performed in TBtools (v2.083) using a one-sided hypergeometric test: $P = 1 - \sum_{i=0}^{m-1} \left(\frac{\binom{M}{i} \binom{N-M}{n-i}}{\binom{N}{n}} \right)$, [54]. In this software, enriched P -values were calculated according to one-sided hypergeometric test: with N representing the number of gene with KEGG annotation, n standing for the number of DEGs in N , M indicating the number of genes in each KEGG term, and m suggesting the number of DEGs in each KEGG term.

Insect bioassays

To survivorship analysis, newly emerged *B. tabaci* adults were microinjected with dsRNA and placed on *N. tabacum* leaves. After 24 h, any dead individuals (potentially due to injection injury) were removed. For each treatment, 20-30 surviving adults were transferred to leaf cages, and mortality was recorded daily for 10 days. Three independent biological replications were performed.

For fecundity analysis, dsRNA-treated virgin females were paired with untreated males in leaf cages. Five such pairs were maintained per cage ($n=10$ -12 cages per treatment). Egg deposition was quantified daily for 3 consecutive days under controlled conditions. Each male-female pair was considered an experimental replicate.

For host preference test, detached *N. tabacum* leaves with moist cotton-wrapped petioles were arranged in a 50-cm diameter choice arena with a central release chamber. Forty adult whiteflies were released per chamber, and settling preference was recorded at 3, 6, 12, 24, 36, and 48 h post-release. The experiment included 10-12 replicates per treatment, with leaf positions randomized between replicates to eliminate positional bias.

EPG recording and analysis

Feeding behavior of *B. tabaci* was monitored using a GiGA-8d EPG amplifier (Wageningen Agricultural University; 10 TΩ input resistance, <1 pA input bias current) with 50× gain and ±5 V output range. Prior to recording, dsRNA-treated whiteflies were starved for 12 h on moist filter paper, briefly anesthetized with CO₂ (10 s), and connected to the amplifier via a gold wire electrode (20 μm diameter × 5 cm length) attached to the pronotum using conductive silver glue. Plant electrodes consisted of copper wires (2 mm diameter × 10 cm length) inserted into the soil of potted *N. tabacum*. Recordings were conducted for 8 h in a Faraday cage (120×75×67 cm; Dianjiang, Shanghai), with only insects surviving the full duration retained for analysis.

Recorded waveforms were processed using PROBE 3.4 software (Wageningen Agricultural University) and categorized into four distinct feeding phases: nonpenetration (np), pathway duration (C), phloem salivation (E1), and phloem ingestion (E2) as defined in a previous study [30]. Each treatment included ≥ 10 biological replicates, with dsGFP-injected and untreated whiteflies serving as controls.

Agrobacterium-mediated plant transformation in *N. tabacum* and *N. benthamiana*

The recombinant expression vectors were introduced into *A. tumefaciens* strain GV3101 via heat shock transformation. Transformed cells were selected on LB agar plates supplemented with 50 μg/mL kanamycin and 10 μg/mL rifampicin, followed by incubation at 28°C for 60 h. Positive colonies were inoculated into liquid LB medium with the same antibiotics and grown to log phase.

Bacterial cells were then pelleted by centrifugation at 2,400 *g* for 2 min and resuspended in induction buffer (10 mM MgCl₂, 10 mM MES [pH 5.6], 200 μM acetosyringone) adjusted to specific OD₆₀₀ values depending on the experiment.

For standard experiments, bacterial suspensions were normalized to OD₆₀₀ = 1.0, while degradation assays used three concentrations (OD₆₀₀ = 0.05, 0.3, and 1.0). Equal volumes of selected bacterial suspensions were mixed and infiltrated into leaves of 4–5-week-old *N. tabacum* and *N. benthamiana* plants using a needleless syringe. Infiltrated plants were maintained under controlled conditions for subsequent analysis.

Diaminobenzidine (DAB) staining

To detect H₂O₂ accumulation in *N. tabacum* leaves, leaf samples were immersed in 1 mg/mL 3,3'-diaminobenzidine tetrahydrochloride (DAB-HCl, #D8001, Sigma-Aldrich, St. Louis, MO, USA) solution (pH 3.8) for 6 hours at room temperature, allowing H₂O₂-dependent brown polymerization to occur. Following incubation, the DAB solution was replaced with 100% ethanol for chlorophyll removal, and samples were decolorized overnight at 65°C. The destained leaves were then photographed under standardized conditions using a Canon EOS 80D digital camera (Canon Inc., Tokyo, Japan).

Interaction assays between two proteins

In the Y2H screening assay, the coding sequence of BtRDP^{-SP} was cloned into the pGBKT7 bait vector (Clontech, USA), while a *N. benthamiana* cDNA library was constructed in the pGADT7 prey vector (Biogene Biotech, Shanghai, China). These recombinant vectors were co-transformed into yeast strain Y2HGold and plated on QDO medium (SD/-Ade/-His/-Leu/-Trp, #630428, Takara) for 3 days at 30°C. Positive colonies were cultured in QDO liquid medium, and plasmids were extracted using a yeast DNA kit (#DP112-02, TIANGEN, Beijing, China) before transformation into *E. coli* DH5α (#TSC-C14, Tsingke) for sequencing (YouKang Biotech, China) to identify interacting partners.

In the pairwise Y2H verification assay, BtRDP^{-SP} and NISP104^{-SP}, and mutant variants of NtRPLP4/SIRLP4/OsRPLP4 were cloned into either pGBKT7 or pGADT7 vectors using primers from Table S4. Given that Y2H is generally difficult to identify membrane receptors, the truncated versions of NtRPLP4/SIRLP4/OsRPLP4 lacking the signal peptide and transmembrane domains were used. Yeast co-transformants were first selected on DDO medium (SD/-Leu/-Trp) (#630417, Takara) for 3 days at 30°C, then spotted onto QDO medium to confirm interactions through growth after another 3 days of incubation.

In the Co-IP assays, Full-length sequences of BtRDP, NISP104, NtSOBIR1, NtRPLP4, SIRLP4, and OsRPLP4 were cloned into lic-myc or lic-flag vectors for C-terminal tagging, with BtFTSP-flag serving as a negative control. Five-week-old *N. benthamiana* leaves were agroinfiltrated with these constructs or collected from oeBtRDP transgenic plants. Total proteins were extracted from 1 g frozen tissue in IP lysis buffer (#87788, Thermo Scientific) with protease inhibitors (#56079200, Roche, Switzerland), followed by centrifugation at 1,000 × *g* for 20 min. Cleared lysates were incubated with anti-flag beads (#L00432-1, GenScript, Nanjing, China) for 4 h at 4°C, washed with PBS, and eluted in 2× SDS-PAGE buffer (500 mM Tris-HCl, pH=6.8, 50% glycerol, 10% SDS, 1% bromophenol blue, and 2% β-mercaptoethanol) for Western-blotting assay.

In the BiFC assay, BtRDP^{-SP}, BtFTSP^{-SP}, NtRPLP4, and NtCf9-like were cloned into the pCV-cYFP or pCV-nYFP vectors, respectively. After transforming *A. tumefaciens* GV3101, pairwise combinations were co-infiltrated into *N. benthamiana* leaves. Following 36–48 h incubation in a growth chamber, reconstituted YFP fluorescence was visualized using a Leica SP8 confocal microscope with standard filter sets for YFP detection.

RLP degradation by salivary effectors among different herbivorous insects

To examine salivary protein effects on RLPs, we co-infiltrated *A. tumefaciens* GV3101 strains carrying various constructs into *N. benthamiana* leaves. The recombinant vectors expressed C-terminally tagged proteins: BtRDP-flag, NISP104-flag, GUS-flag, NISP7-flag, and myc-tagged RLPs (NtRLP4, SiRLP4, OsRLP4, SlCf9, NtCf9). Bacterial suspensions were mixed at predetermined OD₆₀₀ ratios before infiltration, with leaf samples collected 48 hours post-infiltration for subsequent analysis.

For confocal microscopy studies, *N. benthamiana* leaves were co-infiltrated with *A. tumefaciens* carrying: (1) NtRLP4-GFP plus BtRDP-mCherry, (2) NtRLP4-GFP plus RFP-mCherry, or (3) NtRLP4-GFP plus BtFTSP-mCherry. The samples were imaged by confocal microscopy at 48 h post injection. The same parameters were used in photographing.

This study also conducted a treatment on *N. benthamiana* plants that were overexpressing NtRLP4-myc using purified proteins. Briefly, the BtRDP^{−SP} and GFP coding sequences were cloned into pET-28a for 6×His-tagged expression in *E. coli* Transetta (TransGen Biotech). Protein expression was induced with 0.5 mM IPTG at 25°C for 8 h, followed by purification using Ni-NTA agarose and buffer exchange into PBS using a 3-kDa molecular-weight cutoff Amicon Ultra-4 Centrifugal Filter Device (Millipore, MA, USA). Different concentrations of purified proteins were infiltrated into leaves of *N. benthamiana* plants stably expressing NtRLP4-myc, with samples collected 24 h post-treatment.

To investigate degradation pathways, we infiltrated *N. benthamiana* leaves with: 50 μM MG132 (#M7449, Sigma-Aldrich, Steinheim, Germany), 5 μM 3-MA (#M9281, Sigma-Aldrich,), or 10 nM BAF (#BML-CM110-0100, Enzo Life Sciences, Farmingdale, USA) at 24 h post-agroinfiltration. The samples were incubated for additional 24 h. Protein levels were detected using Western-blotting assay.

In planta ubiquitination assay

Ubiquitination detection in *N. benthamiana* was performed according to the methods previously reported [55]. The recombinant vector for transient expression of the HA-fused ubiquitin protein was kindly provided by Dr. Chao Zheng (Ningbo University). Myc-tagged proteins were co-expressed with HA-UBQ by agroinfiltration, and 50 μM MG132 was injected into the *N. benthamiana* leaves one day post *A. tumefaciens* infiltration. Immunoprecipitation was performed using anti-myc beads (ytma-20, ChromoTek, Planegg-Martinsried, Germany). Ubiquitination signals were detected by immunoblot with anti-HA antibody (HT301, TransGen Biotech, Beijing, China).

Scanning electron microscopy (SEM)

B. tabaci adults were allowed to feed on artificial diet solutions for 24 h. Parafilm sections containing salivary sheaths were carefully excised and rinsed with 1× PBS (pH 7.4) to remove residual diet. The samples were mounted on aluminum stubs, desiccated under vacuum, and sputter-coated with gold before imaging with a TM4000 II Plus SEM (Hitachi, Tokyo, Japan). Salivary sheath length (from apex to base) was measured, with 20 sheaths analyzed per treatment to ensure statistical reliability.

Statistical analysis

The log-rank test (SPSS Statistics 19, Chicago, IL, USA) was used to determine the statistical significance of survival distributions. Two-tailed unpaired Student's *t*-test (comparisons between two groups) or one-way ANOVA test followed by Tukey's multiple comparisons test (comparisons among three groups) was used to analyze the results of qRT-PCR, fecundity analysis, and host choice analysis. The EPG data were first checked for the normality and homogeneity of variance,

and those not fitting a normal distribution were $\log_{10}$ transformed before Student's *t* test, as described in previous studies [30]. Statistical tests, including Student's *t*-test and one-way ANOVA, and data visualization were conducted in GraphPad Prism 9.

Data availability

All data is included in the manuscript and/or supporting information. The transcriptomic data have been submitted to the National Genomics Data Center under accession number PRJCA025907. The original blot images have been displayed in Fig. S29-S32.

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Additional information

Author Contributions Statement

H-JH, J-ML, C-XZ, and J-PC planned and designed the research. XW, J-BL, Y-ZW, and H-JH performed experiments and analyzed data. X-HZ provided the EV plants. H-JH and XW drafted the manuscript.

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Additional files

[Supplemental figures and tables](#) 

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Peer reviews

Reviewer #1 (Public review):

Summary:

This manuscript investigates how herbivorous insects, specifically whiteflies and planthoppers, utilize salivary effectors to overcome plant immunity by targeting the RLP4 receptor.

Strengths:

The authors present a strong case for the independent evolution of these effectors and provide compelling evidence for their functional roles.

Comments on revisions:

The authors have addressed all my concerns.

<https://doi.org/10.7554/eLife.108737.3.sa3>

Reviewer #2 (Public review):

Summary:

The authors tested an interesting hypothesis that white flies and planthoppers independently evolved salivary proteins to dampen plant immunity by targeting a receptor like protein. Unlike previously reported receptor like proteins with large ligand-binding domains, the NtRLP4 here has a malectin LRR domain. Interestingly, it also associates with the adaptor SOBIR1. While the function of this protein remains to be further explored, the authors provide strong evidence showing it's the target of salivary proteins as the insects' survival strategy.

The authors have nicely addressed the questions I raised.

I noticed two small points the authors may modify:

- Line 16: delete "on"

- Line 185: Replace "is resistant to *B. tabaci* infestation" with "confers resistance against *B. tabaci*".

<https://doi.org/10.7554/eLife.108737.3.sa2>

Reviewer #3 (Public review):

Summary:

In this study, Wang et al., investigates how herbivorous insects overcome plant receptor-mediated immunity by targeting plant receptor-like proteins. The authors identify two independently evolved salivary effectors, BtRDP in whiteflies and NISP694 in brown planthoppers, that promote the degradation of plant RLP4 through the ubiquitin-dependent proteasome pathway.

Strengths:

This work highlights a convergent evolutionary strategy in distinct insect lineages and advances our understanding of insect-plant coevolution at the molecular level.

Comments on revisions:

The authors have satisfactorily addressed all the issues I raised.

<https://doi.org/10.7554/eLife.108737.3.sa1>

Author response:

The following is the authors' response to the previous reviews

Public Reviews:

Reviewer #1 (Public review):

Summary:

This manuscript investigates how herbivorous insects, specifically whiteflies and planthoppers, utilize salivary effectors to overcome plant immunity by targeting the RLP4 receptor.

Thank you for your comments.

Strengths:

The authors present a strong case for the independent evolution of these effectors and provide compelling evidence for their functional roles.

Thank you for your help in improving our manuscript

Reviewer #2 (Public review):

Summary:

The authors tested an interesting hypothesis that white flies and planthoppers independently evolved salivary proteins to dampen plant immunity by targeting a receptor-like protein. Unlike previously reported receptor-like proteins with large ligand-binding domains, the NtRLP4 here has a malectin LRR domain. Interestingly, it also associates with the adaptor SOBIR1. While the function of this protein remains to be further explored, the authors provide strong evidence showing it's the target of salivary proteins as the insects' survival strategy.

Thank you for your comments.

Major points:

The authors mixed the concepts of LRR-RLPs with malectin LRR-RLPs. These are two different type of receptors. While LRR-RLPs are well studied, little is known about malectin LRR-RLPs. The authors should not simply apply the mode of function of LRR-RLPs to RLP4 which is a malectin LRR-RLP. In addition, LRR-RLPs that function as ligand-binding receptors typically possess >20 LRRs, whereas RLP4 in this work has a rather small ectodomain. It remains unclear whether it will function as a PRR. I can't agree with the author's logic of testing uninfested plants for proving a PRR's function. The function of a pattern recognition receptor depends on perceiving the corresponding ligand. As shown by the data provided, RLP4-OE plants have altered transcriptional profile indicating activated defense, suggesting it's unlikely a PRR. An alternative explanation is needed. More work on BAK1 will also help to clarify the ideas proposed by the authors.

We sincerely thank the reviewer for the insightful and constructive comments, which have helped us critically re-evaluate our interpretation of RLP4 function. In the revised manuscript, we have addressed this important point by adding a detailed discussion of an alternative explanation for RLP4's role in plant defense. Specifically, we now explicitly distinguish between classical LRR-RLPs and malectin-domain-containing RLPs, and we acknowledge that RLP4 may not function as a canonical PRR. We also discuss the structural features of RLP4, including its malectin-like domain and relatively small LRR region, and the observation that NtRLP4 overexpression lines exhibit altered transcriptional profiles even in the absence of insect infestation. Based on these lines of evidence, we propose that RLP4 may instead act as a regulatory component within plant immune signaling networks, modulating defense outputs rather than functioning as a direct ligand receptor. The revised discussion now reads as follows: "Together, this study reveals that suppressing PRR-mediated plant immunity may be a conserved strategy employed by herbivorous insects for successful feeding. We demonstrate that whiteflies and planthoppers have independently evolved salivary effectors that facilitate the ubiquitin-dependent degradation of defensive RLP4 in host plants, thereby dampen RLP4-mediated plant immunity (Fig. 6). Nevertheless, the precise

mechanism by which RLP4 contributes to plant defense warrants further consideration. While it may function as a canonical PRR that perceives insect-derived molecular patterns, several lines of evidence point to an alternative interpretation. Structurally, RLP4 differs from classical LRR-RLP: it contains a malectin-like domain and a relatively small LRR domain, contrasting with typical LRR-RLPs that often possess large LRRs dedicated to ligand binding. Functionally, NtRLP4 overexpression lines exhibit significantly altered transcriptional profiles and dysregulated SA/JA pathways even in the absence of insect infestation, a phenotype inconsistent with canonical PRRs, which typically remain quiescent until ligand perception. These findings point to an alternative explanation: rather than functioning as a classical PRR that recognizes insect-derived molecules, RLP4 may act as a regulatory component within plant immune signaling networks. Elucidating the precise mechanism of RLP4 in conferring plant defense against herbivorous insects will therefore be an important focus of future research” in Line 392-407.

Reviewer #3 (Public review):

Summary:

In this study, Wang et al., investigate how herbivorous insects overcome plant receptor-mediated immunity by targeting plant receptor-like proteins. The authors identify two independently evolved salivary effectors, BtRDP in whiteflies and NISP694 in brown planthoppers, that promote the degradation of plant RLP4 through the ubiquitin-dependent proteasome pathway. NtRLP4 from tobacco and OsRLP4 from rice are shown to confer resistance against herbivores by activating defense signaling, while BtRDP and NISP694 suppress these defenses by destabilizing RLP4 proteins.

Thank you for your comments.

Strengths:

This work highlights a convergent evolutionary strategy in distinct insect lineages and advances our understanding of insect-plant coevolution at the molecular level.

Two minor comments:

In line 140, yeast two-hybrid (Y2H) was used to screen for interacting proteins in plants. However, it is generally difficult to identify membrane receptors using Y2H. Please provide more methodological details to justify this approach, or alternatively, include a discussion explaining this.

Thank you for pointing this out. It is true that Y2H is generally difficult to identify membrane receptors. To address this limitation, we used truncated versions of RLP4s lacking the signal peptide and transmembrane domains in point-to-point Y2H assays. In addition, the interactions between BtRDP and RLP4s were further validated by Co-IP and BiFC experiments. In the revised manuscript, we have clarified this methodological detail as follows: “Given that Y2H is generally difficult to identify membrane receptors, the truncated versions of NtRLP4/SIRLP4/OsRLP4 lacking the signal peptide and transmembrane domains were used” in Linr 636-638.

In Figure S12C, the interaction between the two proteins appears to be present in the nucleus as well. Please provide a possible explanation for this observation.

Thank you for pointing this out. During revision, we further examined the subcellular localization of NtRLP4 and found that NtRLP4-GFP could also be detected in the nucleus when expressed alone (Fig. S18), suggesting that NtRLP4 may have additional functions beyond serving as a cell surface pattern recognition receptor. In the revised manuscript, we discussed that NtRLP4 might play other roles in addition to PRRs in the discussion section as

follow: “Together, this study reveals that suppressing PRR-mediated plant immunity may be a conserved strategy employed by herbivorous insects for successful feeding. We demonstrate that whiteflies and planthoppers have independently evolved salivary effectors that facilitate the ubiquitin-dependent degradation of defensive RLP4 in host plants, thereby dampen RLP4-mediated plant immunity (Fig. 6). Nevertheless, the precise mechanism by which RLP4 contributes to plant defense warrants further consideration. While it may function as a canonical PRR that perceives insect-derived molecular patterns, several lines of evidence point to an alternative interpretation. Structurally, RLP4 differs from classical LRR-RLP: it contains a malectin-like domain and a relatively small LRR domain, contrasting with typical LRR-RLPs that often possess large LRRs dedicated to ligand binding. Functionally, NtRLP4 overexpression lines exhibit significantly altered transcriptional profiles and dysregulated SA/JA pathways even in the absence of insect infestation, a phenotype inconsistent with canonical PRRs, which typically remain quiescent until ligand perception. These findings point to an alternative explanation: rather than functioning as a classical PRR that recognizes insect-derived molecules, RLP4 may act as a regulatory component within plant immune signaling networks. Elucidating the precise mechanism of RLP4 in conferring plant defense against herbivorous insects will therefore be an important focus of future research” in Line 392-407.

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

The authors have addressed all my concerns.

Thank you for your help in improving our manuscript

Reviewer #2 (Recommendations for the authors):

This work is quite interesting. It's not necessary to prove RLP4 as a PRR to show the merit of this discovery. The current logic is forced and thus the conclusion not convincing. Finding an alternative explanation will be more helpful.

Thank you for your valuable suggestions. In the revised version, we discussed the alternative explanation as follow: “Together, this study reveals that suppressing PRR-mediated plant immunity may be a conserved strategy employed by herbivorous insects for successful feeding. We demonstrate that whiteflies and planthoppers have independently evolved salivary effectors that facilitate the ubiquitin-dependent degradation of defensive RLP4 in host plants, thereby dampen RLP4-mediated plant immunity (Fig. 6). Nevertheless, the precise mechanism by which RLP4 contributes to plant defense warrants further consideration. While it may function as a canonical PRR that perceives insect-derived molecular patterns, several lines of evidence point to an alternative interpretation. Structurally, RLP4 differs from classical LRR-RLP: it contains a malectin-like domain and a relatively small LRR domain, contrasting with typical LRR-RLPs that often possess large LRRs dedicated to ligand binding. Functionally, NtRLP4 overexpression lines exhibit significantly altered transcriptional profiles and dysregulated SA/JA pathways even in the absence of insect infestation, a phenotype inconsistent with canonical PRRs, which typically remain quiescent until ligand perception. These findings point to an alternative explanation: rather than functioning as a classical PRR that recognizes insect-derived molecules, RLP4 may act as a regulatory component within plant immune signaling networks. Elucidating the precise mechanism of RLP4 in conferring plant defense against herbivorous insects will therefore be an important focus of future research” in Line 392-407.

Inappropriate descriptions still exist at multiple places across the manuscript and damages the merit of this work. I highly recommend the authors to consult an expert in plant PRR research for proof reading. The language editing service the authors used only provided limited help in this case. Here are a few examples:

We sincerely thank the reviewer for the critical and constructive comments. We agree that precise language is essential for conveying scientific findings. In the revised version, we have refined the text with the help of colleagues who have expertise in plant immunity, aiming to ensure the descriptions are as precise and professional as possible.

Line 16: Using "depend" ignores the fact that many biotic invaders are recognized by NLRs. The authors can simply use the word "use" or "utilize".

Thank you for your suggestion. We corrected it in the revised version.

Line 20: "target defensive RLP4, therefor minimizing the plant immunity" is a strange saying. "dampen RLP4-mediated plant immunity" will be better.

Thank you for your suggestion. We corrected it in the revised version.

Line 49: as far as I know, only LRR-RLPs use SOBIR1 as adaptor. The authors should introduce this specific point. The mode of action of other type of LRR-RLPs are less clear.

Thank you for your suggestion. In the revised version, we re-introduce this as follow: "As RLPs lack the intracellular signaling domains, they are anticipated to associate with adaptor kinases to form the bimolecular receptor kinases. For example, suppressor of BAK1-interacting receptor-like kinase 1 (SOBIR1) is reported to act as a common adaptor for most, if not all, of the leucine-rich repeat RLP (LRR-RLP)" in Line 48-52, "The receptor-like kinase SOBIR1, which contained a kinase domain, has been widely reported to be required for the function of LRR-RLPs in the innate immunity. However, whether SOBIR1 interacted with malectin-LRR RLP remains largely unknown" in Line 170-173.

Line 67: There are quite a few publications showing that insect salivary proteins dampen plant immunity.

Sorry for the inaccurate description. We agree that an accumulated literature describes the suppression of plant immunity by insect salivary proteins. However, the specific molecular mechanism by which these proteins target plant PRRs is still poorly understood. In the revised version, we specified that "it remains largely unknown how insects cope with plant PRRs" in Line 68-69.

Line 149: I don't understand what "point-to-point Y2H" is.

Thank you for your comment. We agree that the term "pairwise Y2H" is more commonly used in the literature than "point-to-point Y2H." To avoid any confusion and to align with standard terminology, we have replaced "point-to-point Y2H" with "pairwise Y2H" throughout the revised manuscript.

Line 179: Replace with "NtRLP4 and NtSOBIR1 confers resistance to B. tabaci". You don't say a protein is resistant to a insect infestation. The same applies for Line 209-210.

Thank you for your suggestion. We corrected it in the revised version.

Minor points:

Line 91-92: Lengthy text for simple results.

Line 98: "which was significantly different from the actin or ribosomal 18S rRNA" can be deleted. It's self-evident that actin and 18S rRNA are controls. The same applies to Line 101.

Line 130: unnecessary sentence, delete.

The use of verb forms needs further correction.

Thank you for your valuable suggestion. In the revised manuscript, we have revised the text accordingly. We truly appreciate your help in improving our manuscript.

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